ADAPTIVE AND EXPLAINABLE HUMAN-ROBOT COLLABORATION FOR INTELLIGENT SORTING SYSTEMS

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DECLARATION

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ABSTRACT

The present study proposes a novel Human–Robot Collaboration (HRC) framework aimed at developing an inclusive and interactive platform for cognitive and sensory learning for differently abled students. The system integrates four key components into a unified architecture to support real-time perception, interaction, and decision-making. The first component is a vision-based texture recognition module built on a self-supervised transformer model (DINOv2), which enables accurate identification of object materials by capturing both local and global visual features. The second component is a robotic learning module designed to enhance pattern recognition and linguistic skills through structured object-based exercises, where robotic actions are used to reinforce learning concepts. The third component introduces a non-invasive head and gaze-tracking system using only an RGB camera, enabling intuitive object selection and human–robot interaction without the need for specialized hardware. The fourth component is an explainable AI (XAI)-driven predictive sorting system that improves transparency in robotic decision-making by providing interpretable insights into classification and action processes. By integrating object detection, robotic manipulation, gaze-based interaction, and speech feedback, the proposed system establishes a multimodal and adaptive learning environment. The human-in-the-loop design allows learners to observe robotic actions, predict outcomes, and receive immediate feedback, thereby enhancing active participation and conceptual understanding. The inclusion of real-time explainability further strengthens user trust and system transparency, making complex AI decisions more accessible to learners and educators.

Experimental findings demonstrate that transformer-based perception models outperform conventional CNN architectures in accuracy and robustness, while multimodal integration significantly improves system reliability under varying conditions. The results also confirm that gaze-guided interaction provides an effective and low-cost alternative to hardware-based tracking systems, and that explainable predictive models enhance interpretability in educational robotics applications. Overall, the system shows strong potential in improving engagement, comprehension, and inclusivity in assistive learning environments through intelligent human–robot collaboration.

Keywords: Human–Robot Collaboration, Explainable Artificial Intelligence, Multimodal Learning, Gaze Tracking, Transformer-Based Vision Models

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TABLE OF CONTENTS

DECLARATION i
ABSTRACT ii
ACKNOWLEDGEMENT iii
TABLE OF CONTENTS iv
LIST OF FIGURES viii
LIST OF TABLES ix
LIST OF ABBREVIATIONS x
1. INTRODUCTION 1
1.1 Background & Literature survey 1
1.1.1 Background 1
1.1.2. Literature Review 4
1.1.2.1. Intelligent Robotic Sorting Systems and Their Evolution 4
1.1.2.2. Human-Robot Collaboration in Education 6
1.1.2.3. Texture-Based Perception and Material Recognition in Robotics ... 6
1.1.2.4. Adaptive Pattern Recognition via Human Behavior in Robotics 7
1.1.2.4.1. Adaptive Behavior Understanding in Collaborative Systems 7
1.1.2.4.2. Vision-Based Perception for Human Interaction 7
1.1.2.4.3. Speech and Natural Language Interaction in Robotics 8
1.1.2.4.4. Multimodal Learning and System Integration 8
1.1.2.5. Gaze-guided robotic assistance 9
1.1.2.5.1. Gaze-Based Interaction for Assistive Applications 9
1.1.2.5.2. Vision-Based Gaze Estimation and Deep Learning Approaches .. 10
1.1.2.5.3. Gaze in Collaborative Robotics and System Integration 10

1.1.2.6. XAI and Recommendation Systems in Robotics 12
1.1.2.6.1. Robotic Perception and Manipulation with Yolo 12
1.1.2.6.2. Explainable Artificial Intelligence (XAI) in Robotics 12
1.1.2.6.3. Recommendation Systems for Robotic Efficiency 13
1.1.2.7. Existing Technologies, Sensors, and Hardware 14
1.1.2.8. Technology-Assisted Learning for Differently-Abled Children 16
1.1.2.9. Challenges in current systems 17
1.2. Research Gap 19
1.3. Research Problem 21
1.4. Research Objectives 23
1.4.1. Main Objective 23
1.4.2. Specific Objectives 24
3. METHODOLOGY 28
3.1. Overview of the Integrated Research System 28
3.2. Research Design 30
3.3. Proposed System Architecture 31
3.3.2. Input and Sensing Layer 31
3.3.3. Preprocessing and Feature Extraction Layer 32
3.3.4. Deep Learning Classification and Estimation Layer 33
3.3.5. Intention Mapping and Reasoning Layer 33
3.3.6. Robotic Control and Interaction Layer 34
3.4. Dataset Development and Data Acquisition 35
3.4.1. Gaze Direction Dataset 35
3.4.2. Educational Object Dataset 36
3.4.3. Material Classification Dataset 37
3.4.4. Sorting Arrangement Dataset 37
3.4.5. Dataset Splitting and Balancing 38

3.5. Data Preprocessing and Augmentation 38
3.6. Vision-Based Perception Modules 40
3.6.1. Gaze Direction Classification Using ResNet-18 40
3.6.2. Object Detection Using YOLO 41
3.6.3. Material Classification Using DINOv2 42
3.7. Spatial Mapping and Scene Understanding 44
3.7.1. Gaze-to-Region Workspace Mapping 44
3.7.2. Homography-Based Scene Coordinate Transformation 45
3.8. Predictive Sorting and Recommendation System 45
3.8.1. Feature Engineering and Data Preparation 46
3.8.2. Random Forest Classifier 46
3.8.3. XGBoost and Logistic Regression Models 47
3.9. Explainable AI Integration 48
3.9.1. SHAP-Based Model Explanation 48
3.9.2. Natural Language Explanation Generation 49
3.9.3. Visual Explainability and Interactive Transparency 49
3.10. Speech Recognition and Verbal Interaction Module 50
3.10.1. Speech-to-Text Processing Using Whisper 50
3.10.2. Tokenization and Response Processing 51
3.10.3. Verbal Response Classification Using Logistic Regression 51
3.10.4. Adaptive Multimodal Feedback Mechanism 52
3.11. Robotic Arm Control Architecture 52
3.11.1. Hardware Components 53
3.11.2. Motion Planning and Trajectory Generation 53
3.11.3. Pick-and-Place Execution Pipeline 54
3.11.4. Interactive Learning Workflow Integration 55
3.12. System Integration and Interaction Workflow 55

3.13. Web-Based Teacher Interface and Support System 58
3.14. Experimental Setup and Performance Evaluation 59
System performance was evaluated through a comprehensive multi-stage testing protocol that assessed each component individually and the complete integrated system collectively. The testing methodology follows established practices in intelligent systems evaluation, combining quantitative performance metrics with qualitative assessments of usability, responsiveness, and educational effectiveness. 59
3.14.1. Gaze Classification Evaluation 59
3.14.2. Object Detection and Material Classification Evaluation 60
3.14.3. Speech Recognition and Classification Evaluation 60
3.14.4. Sorting Recommendation Evaluation 61
3.14.5. Task-Level and System-Level Evaluation 61
3.15. Safety and Reliability Considerations 62
3.16. Commercialization Potential and Applications 63
4. Results and Discussion 66
4.1. Results 66
4.2. Research Findings 70
4.3. Discussion 72
5. Conclusion and Future works 74
6. References 76

LIST OF FIGURES

Figure 1. 1:Figure of Layers of the system 29
Figure 1. 2: Final working system 69

LIST OF TABLES

Table 1: Workflow 56
Table 2: Results and Findings 69

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
COCO	Common Objects in Context
CPU	Central Processing Unit
CSS	Cascading Style Sheets
DINOv2	Self-Supervised Vision Transformer Model (DINOv2)
EDA	Exploratory Data Analysis
EOG	Electrooculography
F1-score	Harmonic Mean of Precision and Recall
Fast R-CNN	Fast Region-based Convolutional Neural Network
Faster R-CNN	Faster Region-based Convolutional Neural Network
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping

Abbreviation	Full Form
GUI	Graphical User Interface
HRC	Human-Robot Collaboration
HTML	HyperText Markup Language
HSV	Hue, Saturation, Value
IoT	Internet of Things
IP	Internet Protocol
LIME	Local Interpretable Model-agnostic Explanations
mAP	Mean Average Precision
ML	Machine Learning
MobileNet	Mobile Neural Network Architecture
NLP	Natural Language Processing
PWM	Pulse Width Modulation
R-CNN	Region-based Convolutional Neural Network
ResNet	Residual Neural Network
RGB	Red, Green, Blue
RGB-D	Red, Green, Blue with Depth

Abbreviation	Full Form
RNN	Recurrent Neural Network
ROI	Region of Interest
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SSD	Single Shot Detector
SUS	System Usability Scale
t-SNE	t-Distributed Stochastic Neighbor Embedding
USB	Universal Serial Bus
VGG16	Visual Geometry Group 16-layer Network
WER	Word Error Rate
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting
YOLO	You Only Look Once

1. INTRODUCTION

1.1 Background & Literature survey

1.1.1 Background

The concept of Human-Robot Collaboration (HRC) has emerged as a revolutionary paradigm in intelligent interactive systems, facilitating collaboration between people and robots in shared spaces. Some of the areas where HRC can be implemented include assistive robotics, smart manufacturing processes, and educational processes. HRC provides immense scope in education, specifically with regards to differently abled students as interactive and adaptable learning is beneficial to these students.

The traditional learning process for differently abled individuals has always depended on materials-based learning along with repetitive processes. Although this method is useful to some extent, it lacks the capability of creating engagement for differently abled learners. With the advent of robots, computer vision, and machine learning techniques, it has now become possible to create intelligent systems that facilitate learning through interactivity and observation in an immersive environment.

Although there have been many improvements in the field, some technical issues still need to be considered. The first one is about real-life perception, especially in scenarios such as sorting based on material type for robots. Since there can be differences in texture, illumination, positioning, and background of objects in real life, it becomes challenging to classify these objects correctly [1], [2]. For this reason, the suggested system adopts a vision-based texture recognition system that uses the DINOv2 algorithm [3], which is a self-supervised vision transformer that can obtain high-level visual features. With the help of ROI segmentation techniques, the system can better recognize materials like plastic, metal, glass, and paper.

A good HRC system is also expected to adapt itself to human behavior as well as to have an interactive capability [4], [5]. In relation to educational applications, this implies that systems which will interactively engage themselves with the learners and adapt themselves based on the capabilities of each one are needed. With this in mind, a robotic learning system aimed at facilitating the learning process of both differently abled and regular students is presented herein. This system employs visual-based object detection in combination with interactive tasks involving speech recognition and pattern recognition. Students are guided through adaptive tasks where the robot provides instructions, evaluates responses, and demonstrates correct solutions when necessary, forming a continuous feedback loop that enhances cognitive and verbal skill development [6].

Intent recognition in humans is yet another significant aspect of efficient HRC. It is natural for humans to communicate their intentions using implicit cues like direction of gaze and head motion, which could be utilized for intuitive interactions [7]. The proposed work presents an innovative method for selecting objects based on gaze and head poses using only one RGB camera. The intention recognition is achieved by incorporating all three factors mentioned above in a single framework that works without any extra hardware [8].

Besides perception and interaction, the aspect of decision-making transparency should be considered, especially for education settings in which interpretation is very crucial [9]. Black-box systems could reduce the user's ability to learn from such systems. To solve this problem, an XAI-based predictive sorting component is introduced into the framework. The framework uses a YOLO-based object detection algorithm to recognize the object features, then a decision layer of machine learning algorithms makes sorting decisions according to the distribution of the recognized objects. It is worth mentioning that the framework provides visualization of the reasoning process of the robots' decision making [10].

In response to the demand for interactive and interpretable educational tools suitable for individuals with different abilities, this study offers an integrated HRC model which leverages various forms of advanced technology in a cohesive system. Through the use of texture analysis, interactive learning interactions, gaze-based intent detection, and decision-making explanations, this proposed framework provides multi-modal learning capabilities with a collaboration aspect between humans and robots.

Though activities like sorting, comparison, and classification of items may seem easy for normal learners, it is not always the case when dealing with learners of different cognitive capabilities. This necessitates an exploration of more innovative ways of making the learning experience easier for all learners. Unlike incorporating the use of robotics in the learning experience by just including the technology itself in the process, the proposed study focuses on integrating intelligent machines into the learning environment by involving them in the learning process. Through robotic manipulation and visual inspection of objects, as well as participation from the learner, this will help in understanding relations like categorization, pattern recognition, and classification of items.

In contrast to other techniques that concentrate more on increasing the efficiency of learning for ordinary students, this paper is motivated by the idea of inclusiveness. The purpose of the research is to cater to the requirements of diverse students with different capabilities using a versatile and interactive interface that accommodates the learning styles of each individual student. Through this process, the research makes an effort to contribute to the development of inclusive education technologies.

1.1.2. Literature Review

1.1.2.1. Intelligent Robotic Sorting Systems and Their Evolution

Sorting mechanisms using robotics technology have experienced tremendous progress due to the introduction of AI techniques into the process, thus increasing efficiency and accuracy of these sorting mechanisms. Robotic sorting systems in modern times use algorithms of object detection that employ neural networks with deep learning, and the YOLO model is one of them that makes it possible to recognize and classify objects almost instantaneously [11].

Apart from object recognition, robotic vision techniques include the use of an RGB-D camera which can provide robotic systems with the ability to sense and analyze both color and depth in the environment [12]. As a result, it allows for the improvement of object localization, orientation detection, and grasp planning. Thus, by integrating robotic arms with cameras of this kind, robots can carry out sorting tasks without any human involvement.

Also, the development of collaborative robots (cobotics) has revolutionized the way in which humans interact with machines. While traditional industrial robots operate in enclosed spaces away from human operators, collaborative robots (cobotics) are developed for safe human robot collaboration. Advanced safety measures including force sensors and real-time monitoring systems are utilized by them for achieving this objective of safe and efficient human-robot collaboration [13], [14]. The combination of the advantages provided by humans and robots through this collaboration includes better decision-making capabilities and endurance among other things.

However, there are a number of challenges involved in this process too. Firstly, it must be noted that robotic systems face the challenge of adaptability in varied environments. It becomes very difficult to achieve generalized results since there are varied kinds of objects to deal with, varied light conditions, and space arrangements. Similarly, performing optimally even under dynamic situations where the objects move around is quite difficult [15], [16], [17]. Furthermore, there is a major issue of lack of explainability in most of the AI-based models.

In special needs education, the importance of these aspects becomes more critical, as differently abled learners may require adaptive solutions. It is vital that robotics in education focus on the implementation of interaction in slow pace, high adaptability, and visual guiding for successful training sessions. Hence, any robotic sorting solution must be specifically designed and adjusted to meet the unique needs of special learners.

One of the important features of robotic sorting systems is their real-time operation ability. For this purpose, robotic devices can utilize IoT technology to collect real-time information and communicate with external components. One of the key components required to make robots more interactive and intelligent in the case under discussion are RGB cameras. The cameras are used to gather information about objects' colors, shape, position, among others, which contributes to spatial mapping [18], [19].

Finally, another important enabler that allows enhancing robots' performance is edge computing. The latter refers to executing data processing operations and decision-making directly within devices rather than relying on distant servers. In the current context, edge computing helps minimize delay times in the execution of robot's actions [20].

Another important development is the use of multimodal input systems, which combine multiple sources of data, such as visual, tactile, and physical feedback. This approach enhances the robot's ability to operate in complex and unstructured environments by providing a more comprehensive understanding of the surroundings. Multimodal systems improve robustness and adaptability, allowing robotic sorters to handle a wider range of scenarios effectively [21].

Importantly, the integration of these advanced technologies opens up new opportunities in the field of education. Robotic sorting systems can be utilized as interactive learning tools, particularly for students with special needs. By providing visual, physical, and responsive feedback, these systems can support experiential learning, improve engagement, and enhance cognitive development. As a result, intelligent robotic systems have the potential to play a transformative role in creating inclusive and adaptive educational environments.

1.1.2.2. Human-Robot Collaboration in Education

1.1.2.3. Texture-Based Perception and Material Recognition in Robotics

Deep learning is now the most popular technology in robotics perception. In particular, object detection using YOLO-based networks is a standard practice thanks to the network's performance and accuracy [22]. Perception based on RGB-D technologies also helps to improve robots' functions by adding additional information about object shape and helping to grasp objects [23].

CNN architecture was mostly employed in material recognition and sorting. Nevertheless, according to Zhang et al. [24], conventional deep learning algorithms are unable to

achieve adequate generalization in such environments despite their success in other cases. For that reason, the transformer-based models have been applied more often lately.

Self-supervised learning has made significant progress in visual representation learning. Caron et al. [25] have developed vision transformers able to learn informative features without any labeling process. As for further developments, DINOv2 showed excellent results in fine-grained visual comprehension, which makes it a perfect choice for texture and material recognition and sorting [26].

1.1.2.4. Adaptive Pattern Recognition via Human Behavior in Robotics

1.1.2.4.1. Adaptive Behavior Understanding in Collaborative Systems

One of the major developments in HRC is the robot's capability to learn from human behavior and adjust its responses to human actions [27], [28]. Adaptive pattern recognition further augments this capability by helping to interpret behavioral cues and the surrounding environment in real time [29]. This type of system relies on principles of general artificial intelligence, which involve perceptual capabilities and action toward an objective [30], and human-centered artificial intelligence, which ensures alignment with human objectives [31].

1.1.2.4.2. Vision-Based Perception for Human Interaction

Vision systems are fundamental for enabling real-time understanding in HRC. Early approaches relied on handcrafted features, but deep learning significantly improved object

detection and recognition performance. CNNs demonstrated strong feature extraction capabilities [32], while architectures like ResNet improved depth and accuracy [33].

Region-based methods such as Fast R-CNN and Faster R-CNN further enhanced detection performance through efficient region proposals [34],[35]. For real-time applications, YOLO enables fast object detection in a single forward pass, making it suitable for interactive systems [36],[37].

Additionally, pose estimation frameworks like OpenPose allow robots to interpret human body movements, improving behavioral understanding and interaction quality.

1.1.2.4.3. Speech and Natural Language Interaction in Robotics

Speech communication improves HRC accessibility and provides natural communication. Research on speech recognition and natural language processing can be employed to turn spoken language into structured information ready for computations [38].

Logistic regression algorithms are often applied to classification problems [39],[40], whereas recurrent neural network architectures are used to analyze time-dependent audio streams [41]. These advances help provide more natural interaction between humans and robots.

1.1.2.4.4. Multimodal Learning and System Integration

The combination of multimodal learning, involving visual, kinesthetic, and linguistic engagement, has proved to be effective in improving the efficiency of learning because it involves using various sensory pathways. This technique is highly useful in special education programs since students require different ways of learning.

System-level considerations in collaborative robotics include task allocation and motion planning. Predictive modeling, for example, Markov models, enables predicting human actions [42], while motion planning increases predictability and thus enhances trust [43]. Task allocation can also enhance collaboration efficiency [44]

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1.1.2.5. Gaze-guided robotic assistance

1.1.2.5.1. Gaze-Based Interaction for Assistive Applications

There have been numerous approaches for natural human-to-machine interaction in recent years. Gaze tracking has attracted substantial interest because of its close connection to human attention and intention. In normal human interaction, gaze plays an important role in directing attention, coordination, and intention communication. Gaze becomes an important means of interaction between human and robot [45].

The use of eye movements to communicate intentions offers substantial advantages in assistive applications. Users suffering from physical disabilities are often unable to operate machines using standard methods of controlling the device; however, eye movements alone would suffice to convey user intentions [46], [47].

Early gaze tracking technology depended on specific tools like infrared eye trackers, head-mounted equipment, and electrooculography sensors. Though they were highly accurate, such systems proved to be costly, needed a special environment, and required complex calibration procedures [48], [49].

1.1.2.5.2. Vision-Based Gaze Estimation and Deep Learning Approaches

To overcome the limitations of hardware-based solutions, recent research has shifted toward vision-based gaze estimation using standard RGB cameras. Advances in computer vision and deep learning have enabled the development of methods such as appearance-based gaze estimation, facial landmark detection, and head pose estimation, allowing gaze direction to be predicted using commonly available devices [50], [51].

Deep learning techniques, particularly Convolutional Neural Networks (CNNs), have significantly improved gaze estimation performance. Architectures such as ResNet are effective in extracting spatial features, while lightweight models like MobileNet support efficient real-time inference [52], [53]. Additionally, feature extraction frameworks such as Caffe have been utilized to analyze eye-region characteristics for gaze prediction.

Furthermore, gaze estimation approaches can be broadly categorized into continuous and discrete methods. While continuous methods estimate precise gaze coordinates, discrete methods classify gaze into predefined regions such as left, right, center, up, or down. For robotic applications such as object selection, discrete classification is often more practical as it focuses on intention detection rather than exact gaze position [54], [55].

1.1.2.5.3. Gaze in Collaborative Robotics and System Integration

The use of gaze information in collaborative robots has led to the development of new technologies in intention recognition, situational awareness, and task coordination. With the aid of gaze data, robots can be able to recognize where the human looks at and predict future actions, therefore increasing the efficiency of HRC systems [56], [57].

Nevertheless, existing approaches rely on gaze information either as context or complementary information but not directly as control. Although intention recognition based on gaze is one common application, few have explored the potential of gaze information as an explicit control signal to direct robots to execute certain tasks.

In this paper, we propose a gaze-mediated HRC solution that incorporates gaze estimation, spatial region selection, and object manipulation in a single perception to action framework. Herein, humans can choose objects through selecting particular spatial regions with the use of gaze signals. After obtaining the gaze vector through vision-based gaze estimation, a specific spatial region will be selected. Finally, the robot selects and manipulates the desired object within the chosen region.

The integration of these components also leads to contributions to various areas of research. In assistive robotics, it provides a means of manipulating objects through gaze, increasing autonomy [58]. In inclusive human-computer interaction, it provides a proof-of-concept of an interface based on natural human action, not typical control schemes. Lastly, it contributes to vision-based human-robot collaboration systems in that it integrates intention-aware interaction models [59].

Furthermore, this framework highlights the importance of cost-effectiveness in that it uses standard vision devices and software-only techniques without requiring additional hardware. It also captures a recent trend in computer science that combines computer vision, deep learning, spatial reasoning, and robotics in a single assistive system.

However, the development of such a system entails numerous technical issues related to varying lighting conditions, head postures, and user actions, accurate gaze estimation, gaze-to-spatial region mapping, and proper control of the robotic manipulator. These issues form the basis of the current study that seeks to build a full gaze-based object selection system using a collaborative robotic arm.

1.1.2.6. XAI and Recommendation Systems in Robotics

1.1.2.6.1. Robotic Perception and Manipulation with Yolo

Robotics sorters require a strong capability in terms of perception and manipulation. Some deep learning object detection models such as You Only Look Once (YOLO), Faster R-CNN, and SSD are popularly applied in real-time object detection and classification tasks [60]. Among these models, YOLO becomes more significant since it features high-speed, one-shot detection and is quite applicable in robotic sortation with strict latency requirements.

In order to obtain a better understanding of space, current robotic applications adopt RGB-D sensors to detect object positions to facilitate grasping and collision-free motion planning [61]. In addition, multimodal methods that leverage vision and touch sensors can increase grasping success rates because they help robotic arms recognize the texture, rigidity, and mass of the objects [62].

1.1.2.6.2. Explainable Artificial Intelligence (XAI) in Robotics

XAI is important to enhance the level of transparency and accountability in the robotic system. While models such as YOLO perform very well in perceptual tasks, they act as a black box system making it difficult for us to understand the reasoning process involved in human–robot interaction [63].

The use of various XAI techniques such as LIME, SHAP, and Grad-CAM helps in addressing the above concern. SHAP uses game-theory based contribution of features while Grad-CAM shows the attentive regions that affect image prediction [64], [65].

These can be used in sorting applications to justify the classification of objects as plastic, metal, and glass.

It has been shown that the application of SHAP technique to detect object using YOLO increases the real-time interpretability of the robot vision systems [66]. It helps us to debug, improve trust, and understand decisions made by the robotic system. In decision-making systems, hybrid models that involve decision tree-based approaches alongside machine learning help enhance explainability and reliability [67], [68].

However, the currently available XAI models suffer from the problem of slow real-time performance and poor decision-level interpretation [69], [70].

1.1.2.6.3. Recommendation Systems for Robotic Efficiency

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Recommendation systems in robotics enable intelligent decision-making by recommending the most efficient actions and sequences [71]. The recommendation systems for robotic sorting systems determine the most effective sequence for handling objects, depending on their properties such as shape, size, and material type.

The use of recommendation systems in industry is widespread in such sectors as recycling and warehouse robotics [72], [73]. Recommendation systems in industrial robotics analyze data from various sensors to prioritize the handling of objects and optimize motion complexity.

In this research, recommendation systems will be implemented in a robotized learning environment to make decisions depending on the properties of objects, such as color and shape. Consequently, it is possible to increase the efficiency of operations while

simultaneously contributing to students' learning about classification and pattern recognition.

In robotics, recommendation systems are utilized in combination with multiple sensors for vision, depth, and weight detection, which enables real-time analysis of object distribution and determination of the most efficient sequence for object sorting [74]. A hybrid recommendation system that combines rule-based and machine learning algorithms can enhance both interpretability and flexibility [75]. Nevertheless, real-time recommendations face challenges related to their adaptation to changes in the environment and explanations for recommendation systems' decisions [76].

1.1.2.7. Existing Technologies, Sensors, and Hardware

Current intelligent robotic systems rely on the integration of various sensor technologies, hardware, and artificial intelligence algorithms that enable perception, decision making, and human–robot interaction capabilities. Such robotic systems usually integrate several elements, including visual perception, texture/pattern recognition, gaze-based interaction, explainable decision making, and recommendation systems.

The perception layer relies extensively on visual sensors such as RGB cameras and RGB-D cameras. This visual perception system helps robots detect objects in their vicinity, recognize them, and understand their spatial characteristics. Texture and pattern recognition play a significant role in distinguishing objects based on their physical properties when there are similarities between them.

In addition to vision-related tasks, gaze-based interaction approaches are applied in assistive and collaborative robotics. The use of software-based eye-tracking systems and

gaze estimation algorithms allows the robot to perceive user attention and intentions by identifying the region in space to which the user is looking. As a result, the interaction with the robot becomes more intuitive since users can select and control the object with their eyes.

Explainable Artificial Intelligence methods are included in robot design to increase the transparency of decision-making processes. Feature attribution and other visualization techniques help the system explain its decisions related to object detection and classification. It is especially necessary for collaborative tasks when the human user needs to know the reason why the robot made a certain decision.

As in the case of vision, recommendation approaches are vital in robotic systems. Recommendation algorithms analyze various features of the environment and objects to generate optimal sequences of actions. Robotic sorting systems benefit from recommendations since object prioritization can be based on frequency, categorization, and other criteria.

Robotic systems at a hardware level utilize collaborative robotic arms along with compliance control and safety measures, which allow direct human interaction with the machine. This hardware is also paired up with adaptable grippers and end effectors that can handle a variety of object geometries and material properties. With regard to computational processes, edge computing platforms and specialized AI chips enable timely execution of deep learning algorithms, including perception, XAI and decision modules.

Overall, the combination of a perception module based on visual texture and pattern recognition, gaze direction estimation system, XAI tools, and recommendation-based decision support system, powered by robotics hardware and edge computing systems, constitutes a framework upon which contemporary intelligent robots operate.

1.1.2.8. Technology-Assisted Learning for Differently-Abled Children

There is considerable evidence of the immense value of integrating technology into learning, especially when it comes to children with various learning challenges.

Unlike traditional teaching methods that rely heavily on verbal instruction and manual interaction, modern assistive learning systems integrate artificial intelligence, robotics, and multimodal sensing to create inclusive and engaging learning environments.

A key component of such systems is vision-based perception, where robots use object detection, texture recognition, and pattern analysis to support learning activities such as sorting, classification, and comparison tasks. These capabilities allow learners to understand concepts through visual demonstration and repetition, which is especially beneficial for children who learn better through interactive and sensory-based methods.

Moreover, gaze interaction is essential in enabling accessibility for users with disabilities. With eye-gaze estimation, motor-disabled kids will interact with robots by merely using their eyes without having to use any physical device or even speaking. With the help of tracking their gaze directions, the robotic system will be able to determine which objects the learners are looking at, allowing the learners to manipulate those objects and control robot behavior just by focusing on them.

Recommendation engines are employed in the robotic educational platform as a tool for improving learning processes. The recommendation engine will analyze the learners' behaviors, characteristics of objects, and performance to recommend the optimal learning sequence. The objects may be categorized and displayed according to certain traits like color, shape, or occurrence, which helps the learners develop skills like pattern recognition and decision-making.

Furthermore, XAI is necessary within an education setting because it provides transparency within the robotic decision-making processes. With XAI, there can be clarity within the decision that the robot makes, such as choosing one object among another or recommending a certain group for certain tasks, hence building trust between the learners, educators, and the robots as well as facilitating the learning process.

By integrating all the four aspects of vision-based perception (pattern and texture), gaze-based interaction, recommendation systems, and XAI, a formidable method for assistance in education can be achieved. With this framework, robots will not only serve as devices of automation, but as intelligent companions who can help in educating differently-abled individuals.

1.1.2.9. Challenges in current systems

Despite the impressive developments in robot sorting systems, several limitations still prevail. To begin with, adaptability to changing environments poses a serious challenge. Real-world sorting is often characterized by diverse object geometries, occlusions, and environmental conditions. Although deep learning algorithms are extremely accurate in regularized environments, their performance is compromised in dynamic or new situations unless they are retrained. Agile methods such as quick model adaptation and human-corrected correction have been proposed to address this issue [8]. Indeed, experiments show that human interaction, i.e., occlusion removal or grip pose adaptation, can increase sorting accuracy from ~33% to ~100%, showing the benefit of hybrid collaboration [15]. In the field of education, this flexibility is all the more important, since dealing with students adds another layer of uncertainty.

Secondly, inexplicability of robotic decision-making demoralizes trust and transparency. The majority of sorting robots are "black boxes," where human beings can't comprehend why specific objects were sorted or treated in specific way. Explainable AI (XAI)

frameworks are therefore required to make the decision-making process clear so that human workers can build up trust and work effectively alongside robots [14]. When it comes to educational institutions, transparency becomes even more vital since it would allow educators to comprehend the way the system operates and teach students through its means.

Thirdly, optimization of efficiency still remains challenging where there are multiple kinds of objects. Bots find it hard to optimize the prioritization of objects or acquire task strategies in complex situations. Furthermore, methods based on data such as reinforcement learning, as promising as they are, have limitations such as sample inefficiency, sparse rewards, and sim-to-real gap [13]. Safety and ergonomics concerns also remain: cobots tend to be slower than traditional robots due to safety constraints, which could impair throughput if addressed through sophisticated control and scheduling algorithms [11]. These limitations play an important role within the academic setting, although they could influence the responsiveness and interaction capabilities of the robots if not adequately addressed. Thus, solving these issues is vital when creating effective and functional robots which can be used both in industries and academia.

1.2. Research Gap

Notwithstanding considerable developments in HRC, robotic perception, assistive robotics, and XAI, there are a number of crucial gaps that still remain in existing research where these fields intersect.

Most current HRC robots and robotic applications aim at improving the efficiency, safety, and functionality of the process rather than focusing on visual comprehension or multimodal decision-making in real-world interactive settings. While state-of-the-art deep learning algorithms like YOLO can be used to detect objects in real time, these models are tailored towards object recognition and fail to accommodate material discrimination and pattern comprehension necessary for advanced sorting and assistive robotics applications.

On a similar note, sophisticated visual recognition approaches including self-supervised learning and transformer architectures (DINOv2) show promising representation learning abilities. Yet, their application to real-time robotic systems and multimodal sensor data within HRC frameworks still lacks adequate exploration. The same applies to multimodal sensor fusion (visual, depth, and tactile sensors), which enables more reliable perception but fails to incorporate intelligent decision-making processes.

From the interaction standpoint, gaze-control technologies, recommendation-driven decision models, and XAI solutions are generally developed separately. The gaze-controlled interactions are mostly employed for intention recognition but not for robotic control itself, recommendation systems are used mostly for efficiency optimization and not for knowledge acquisition, and explainable AI is primarily employed after the decision making. This makes the gap between integrated solutions with perception, intention analysis, decision-making, and explanations becoming apparent.

In addition, the vast majority of current assistive and educational robotic solutions tend to specialize exclusively either in behavioral interaction or in perception-related applications. Little attention has been paid to the development of technologies that would combine intelligent perception (texture, pattern, and object recognition), gaze-controlled interaction, decision-making based on recommendations, and explainability within one system.

Yet another significant problem is associated with scalability and affordability issues. For many assistive robotic applications, there is a need to use expensive equipment or individual high-performing modules, which makes the application of such solutions hard to apply within actual educational settings. In addition, the problem of real-time processing imposes limitations on integrating complex models, multi-modal sensing capabilities, and explainable AI within one solution.

Thus, the research gap that can be identified here is related to the development of an effective, real-time, and inexpensive human-robot collaboration approach incorporating YOLO-based perception, pattern recognition and understanding, gaze-based interactions, decision making based on recommendations, and explainable AI.

1.3. Research Problem

While remarkable progress has been made in the areas of HRC, robotic perception, assistive technology, gaze-based interaction, and XAI, there still exists a crucial issue in designing an integrated and intelligent robot system capable of simultaneous real-time perception, decision making, reasoning, and assistive interactions in educational settings.

The current designs for HRC have mostly targeted industrial operations where the primary considerations were safety, productivity, and coordination among different actors. While such designs facilitate systematic cooperation between human operators and robots, they lack adequate capacity for detailed perception and intelligent understanding of objects in actual settings, a prerequisite for assistive learning activities. In the same way, robotic perception techniques relying on deep learning algorithms offer efficient object detection mechanisms but are incapable of capturing subtle differences in textures and materials necessary for accurate classification.

Recent developments in the fields of self-supervised learning and vision transformers have proven effective in terms of achieving transferable visual representations. Nonetheless, the application of these concepts in real-time robotic applications, especially those involving multimodal perception, interaction, and reasoning, is yet to be fully achieved. Furthermore, methods of sensor fusion that improve robustness are increasingly being applied; nonetheless, most are done in loosely-coupled manners without integrating well with learning reasoning systems.

Moreover, robots have proven to be effective in helping achieve better engagement and learning among differentially-abled students through assistive and educational applications. However, most of these applications focus more on behavioral interactions than understanding objects and supporting intelligent sorting.

When considered from the viewpoint of interaction, it appears that gaze detection, recommendation algorithms, and explainable AI technologies are developed separately. Gaze is used primarily to estimate the attention of the user, whereas recommendation techniques are utilized to optimize the efficiency of robotic decision making rather than to help humans make informed educational choices. Moreover, explainable AI algorithms are applied to post-hoc explanation of decisions made by robots and cannot provide real-time explanations.

Thereby, there is an urgent need to develop an algorithm that would be able to integrate all stages from perceiving intentions of users to providing recommendations for robot behavior and explanations of its actions. Moreover, there are no existing approaches for applying recommendation algorithms to educational robotics, while explainable AI technologies have been mainly developed for use in offline environments.

Thus, the primary research problem that this paper investigates is:

What kind of integrated human-robot collaborative system can be developed that incorporates real-time object perception (with deep visual interpretation of textures and patterns), multi-modal sensing, gaze-directed interaction, intelligent decision making through recommendation, and explainable artificial intelligence to ensure transparency and efficient learning assistance for differently abled children?

The research problem underscores the necessity of overcoming the differences between:

- Visual perception techniques (object, texture, and pattern recognition)
- Multi-modal sensing integration
- Human intention identification via gaze tracking
- Intelligent decision making based on recommendations
- Explainable artificial intelligence

1.4. Research Objectives

The rationale behind the current research lies in the holistic vision of creating a human-robot collaborative intelligent system, which would combine the advantages offered by sophisticated robotic platforms, machine perception technologies, explainable artificial intelligence, and inclusive education. The proposed solution aims at becoming an assistive platform for people with special needs that could engage in various activities via gaze-controlled actions, object identification, materials analysis, pattern learning, voice communication, and explainable machine-based decisions. As a whole, the combination of the four proposed solutions helps solve different problems, yet the ultimate goal remains the same – accessibility and education via robots.

1.4.1. Main Objective

The main goal of the proposed research project is the development and validation of an advanced multimodal human-robot collaboration platform that provides support for differently abled persons via intelligent, inclusive learning. It consists of four interrelated modules: (1) a novel gaze-based interaction system that allows users to select objects and manipulate a robot using their eyes; (2) an adaptive teaching robot that performs online texture classification in self-supervised mode utilizing deep learning techniques and multimodal sensor fusion; (3) an interactive learning module that promotes development of pattern recognition abilities and speech communication skills using visual, tactile, and auditory inputs; and (4) an explainable AI-based robotic manipulation and sorting system.

1.4.2. Specific Objectives

To accomplish the main objective, this research is organized around the following specific objectives, which span the four integrated components of the system:

A. Perception and Object Recognition

1. To design and develop a real-time gaze direction estimation model using computer vision and deep learning techniques that operates entirely through software-based approaches without specialized hardware, and classifies user gaze into predefined spatial regions to serve as an intuitive interface for robotic control.
2. To design and develop a real-time object detection system using the YOLO algorithm capable of accurately identifying and localizing predefined objects within the learning and sorting environment, and to create a custom annotated dataset for training and evaluating the detection model under realistic conditions.
3. To develop a vision-based material texture recognition system using deep learning architectures, including Sequential CNN, EfficientNet, MobileNet, and DINOv2, for classifying everyday materials such as plastic, glass, rubber, and metal under varying lighting, orientation, and background conditions, and to evaluate these models comparatively to identify the most accurate and generalizable architecture.

B. Multimodal Sensing and Sensor Fusion

1. To develop a multimodal sensor fusion mechanism that integrates RGB vision-based classification with physical sensing modalities, including ultrasonic sensing for object detection and environmental awareness, in order to improve system robustness and reduce misclassification under real-world operating conditions.

2. To devise a region-based mapping model that associates detected gaze areas with corresponding object positions within the robot's collaborative workspace, thereby translating gaze-based user intentions into spatial commands for robotic manipulation.

C. Robotic Control and System Integration

1. To implement a robotic arm control system using a 6-degree-of-freedom manipulator interfaced with an Arduino microcontroller, capable of receiving and executing commands from the gaze estimation, object recognition, and AI recommendation modules, and to design a complete end-to-end pipeline where the robot identifies, approaches, and grasps target objects based on user intent.
2. To develop a real-time robotic sorting integration module that maps detected object positions onto predefined workspace grids, enabling the robotic arm to execute sequential sorting tasks directed by machine learning model recommendations, using color-coded or shape-defined bins as sorting targets.
3. To develop a pattern-based learning mechanism that enables students to observe and replicate object sequences demonstrated by the robotic arm, thereby reinforcing cognitive skill development and pattern recognition through interactive physical engagement.

D. Intelligent Decision-Making and Explainable AI

1. To design and implement a recommendation-based sorting strategy using a Random Forest classifier to predict optimal object manipulation sequences based on attributes including object count, spatial distance, and reachability, and to compare multiple machine learning models to determine the most effective decision-making approach for real-time robotic guidance.

2. To integrate SHAP (SHapley Additive exPlanations)-based Explainable AI (XAI) into the recommendation system to provide transparent, human-interpretable explanations of sorting decisions in both visual and textual formats, thereby improving educator trust and supporting meaningful human-robot interaction in educational environments.

E. Speech-Based and Multimodal Interaction

1. To integrate a speech-based interaction module that converts visual object patterns into verbal instructions and auditory feedback, supporting language development and communication skill building in students with special educational needs.
2. To apply a logistic regression model for voice command analysis, enabling the system to evaluate the accuracy and relevance of spoken user inputs and provide corrective or confirmatory feedback within the interactive learning loop.
3. To establish a unified multimodal learning framework that coherently combines visual, physical, verbal, and gaze-based interaction modalities, grounded in a constructivist learning philosophy, and to assess its effectiveness in improving student engagement, cognitive abilities, and verbal communication skills in special needs educational settings.

F. Assistive Education and Accessibility

1. To design the integrated robotic system as an inclusive educational platform that enables differently-abled students to observe, interact with, and learn from real-time robotic demonstrations, fostering understanding of material properties, object relationships, and sequential reasoning through hands-on, robot-assisted activities.
2. To develop cost-effective, hardware-accessible implementations of gaze-based control and vision-based perception that remove barriers associated with

specialized or expensive assistive technologies, thereby contributing to the democratization of assistive robotic solutions for individuals with differing physical capabilities.

G. System Evaluation and Validation

1. To experimentally evaluate the complete integrated system with respect to gaze classification accuracy, object detection performance, material recognition precision, robotic task execution reliability, and AI recommendation correctness, using quantitative benchmarks and real-world testing scenarios.
2. To conduct usability testing and educator feedback analysis to assess the interpretability, user-friendliness, and practical effectiveness of the system, including evaluation of the XAI explanations, speech interaction accuracy, and overall system responsiveness in collaborative human-robot scenarios.

3. METHODOLOGY

3.1. Overview of the Integrated Research System

This chapter presents the comprehensive methodology that underpins the design, development, and evaluation of an integrated human-robot collaboration platform. The system described herein represents a convergence of four tightly interrelated research components: (1) a gaze-guided assistive robotic object selection system for differently-abled users, (2) an interactive human-robot collaborative learning system designed for special needs education, (3) an adaptive educational platform centered on intelligent material texture classification and robotic sorting, and (4) an explainable AI-enabled predictive robotic sorting system. Although each component addresses a distinct functional domain, they collectively form a unified, end-to-end intelligent collaboration pipeline that bridges perception, reasoning, explanation, and physical action within shared human-robot environments.

The overarching objective of this research is to establish a modular, scalable, and accessible framework that enables humans to interact with robotic systems through natural modalities, including gaze direction, verbal communication, and physical observation. The integration of these modalities into a single coherent system eliminates the reliance on specialized or expensive hardware, instead leveraging affordable computer vision, deep learning, sensor fusion, and speech processing technologies to realize robust and meaningful human-robot interaction. At its core, the system aspires to serve as both an intelligent assistive device for users with motor or communication impairments and an inclusive educational tool for learners with special needs.

The integrated system operates across three fundamental functional layers that span all four components. The Perception Layer is responsible for acquiring and interpreting sensory data from the environment, encompassing visual input through RGB cameras, gaze estimation through facial analysis, speech input through microphone capture, object

detection through deep learning models, material classification through vision transformers, and proximity sensing through ultrasonic sensors. The Reasoning and Decision-Making Layer synthesizes the outputs of the perception layer into structured decisions and intentions. This includes classifying gaze direction to determine user intent, mapping spatial regions to robotic workspace coordinates, generating learning sequences for educational interaction, predicting sorting strategies through machine learning, and producing natural-language explanations of decisions through Explainable AI techniques. The Action and Interaction Layer translates decisions into physical and communicative outputs, encompassing robotic pick-and-place operations, adaptive verbal and visual feedback, speech synthesis, real-time web-based teacher interfaces, and multimodal reinforcement of learning outcomes.

Each of these layers operates in a coordinated and time-synchronized manner, ensuring seamless transitions between perception, reasoning, and action across all four integrated components. The methodology detailed in this chapter covers every aspect of system development from research design and data acquisition through model training, robotic control architecture, system integration, experimental evaluation, safety considerations, and commercialization potential.

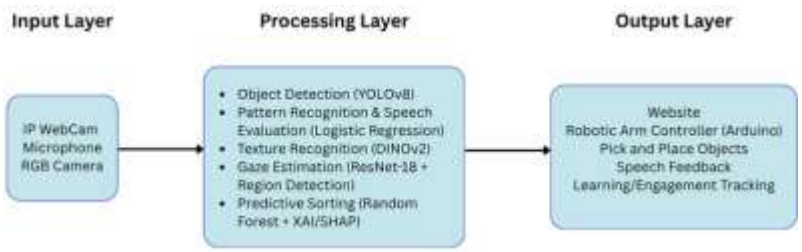

Figure 1. 1:Figure of Layers of the system

3.2. Research Design

This research adopts an experimental system development methodology, characterized by the iterative integration of prototype construction with empirical performance testing. The methodology follows an established research cycle of design, develop, test, and refine, applied across each of the four system components and the integrated system as a whole. This approach is consistent with best practices in human-computer interaction, educational technology research, and intelligent systems engineering, where the complexity and context-dependency of the problem domain necessitate continuous adaptation and improvement grounded in empirical evidence.

The research design is structured around a modular architecture that permits independent development, testing, and optimization of each component while ensuring that inter-component communication and data flow are maintained through well-defined interfaces. This modularity is not merely a technical convenience but a deliberate methodological choice that allows the research team to isolate the contributions of individual components, compare alternative approaches within each module, and progressively integrate validated components into the complete system. It also facilitates troubleshooting and maintenance, as issues can be localized to specific modules without cascading effects on the overall system.

Stakeholder consultation formed a critical component of the research design process. Structured discussions and observational sessions were conducted with educators, therapists, and end users at SERD Kelaniya Special Needs School and associated rehabilitation facilities. These consultations identified key pedagogical priorities, user capability profiles, practical classroom constraints, and safety requirements that directly informed the system design. Specifically, stakeholders emphasized the importance of immediate and specific feedback, multisensory engagement, adaptability to diverse ability levels, and ease of use by teachers with limited technical training. These requirements were translated into concrete design parameters that guided decisions across all four system components.

The research design is organized around three major interconnected components, each encompassing multiple functional modules:

- Vision-Based Perception and Classification Module, encompassing gaze estimation, object detection, and material classification.
- Intention Mapping and Decision-Making Module, encompassing gaze-to-region mapping, predictive sorting recommendation, and Explainable AI integration.
- Robotic Manipulation and Interaction Module, encompassing motion planning, pick-and-place execution, speech-based interaction, and adaptive feedback.

3.3. Proposed System Architecture

The architecture of the proposed integrated system establishes a complete end-to-end pipeline that begins with real-world sensory data acquisition and culminates in intelligent, context-appropriate physical and communicative action. The architecture is designed according to a hierarchical and modular structure, where each layer of the hierarchy abstracts the complexity of the layers below it and presents a simplified interface to the layers above. This design promotes scalability, flexibility, and ease of integration across all four research components.

3.3.2. Input and Sensing Layer

At the base of the architecture lies the input and sensing layer, which is responsible for acquiring raw data from the physical environment through multiple sensing modalities. Visual data is acquired through RGB cameras, implemented using both dedicated hardware cameras and IP Webcam-based

mobile device streaming to provide a cost-effective and flexible imaging solution. These cameras continuously capture video frames of the workspace and the user, providing the visual foundation for gaze estimation, object detection, and material classification. Acoustic data is acquired through high-quality unidirectional microphones positioned to maximize speech clarity while minimizing ambient noise pickup. Proximity and presence data is acquired through ultrasonic sensors that measure the distance to objects within the robotic workspace, providing environmental awareness and object presence validation that complements vision-based perception.

3.3.3. Preprocessing and Feature Extraction Layer

The raw sensory data acquired by the input layer is forwarded to the preprocessing and feature extraction layer, where it is transformed into structured representations suitable for machine learning analysis. Visual data undergoes a multi-stage preprocessing pipeline that includes image resizing to standardized input dimensions, pixel value normalization to the range expected by the respective deep learning models, noise reduction to minimize distortions caused by lighting and sensor artifacts, and face region alignment and cropping for gaze-specific processing. Audio data undergoes noise reduction, silence removal, and volume normalization before being passed to the speech recognition model. These preprocessing operations are critical for ensuring input consistency and optimizing model performance across diverse real-world conditions.

3.3.4. Deep Learning Classification and Estimation Layer

The preprocessed data streams are forwarded to the deep learning classification and estimation layer, which houses the core intelligent perception capabilities of the integrated system. This layer encompasses four primary analytical functions. The Gaze Direction Classifier, implemented using a ResNet-18 convolutional neural network, analyzes facial imagery to classify the user's gaze direction into one of seven predefined spatial regions corresponding to workspace areas. The Object Detection Module, implemented using the YOLO (You Only Look Once) deep learning architecture, processes workspace video frames in real time to detect and localize objects by class and spatial position. The Material Classification Module, implemented using the DINOv2 self-supervised vision transformer, analyzes object images to classify their material composition into predefined categories. The Speech Recognition Module, implemented using OpenAI's Whisper model, processes audio input to accurately transcribe the user's verbal responses into structured text under diverse acoustic conditions. These analytical functions operate concurrently or sequentially depending on the active system mode, with their outputs feeding into the higher layers of the architecture.

3.3.5. Intention Mapping and Reasoning Layer

The outputs of the deep learning layer are consumed by the intention mapping and reasoning layer, which is responsible for transforming raw classification results into structured decisions and action commands. Within this layer, the gaze region predictions are mapped to corresponding workspace coordinates and object targets using a predefined spatial mapping scheme. Object detection results are used to generate structured learning sequences and sort recommendations. Material classification results are integrated with sensor validation data through a decision fusion mechanism that ensures actions are only

initiated when valid and reliable inputs are detected. The predictive sorting recommendation sub-module employs machine learning models, primarily a Random Forest classifier enhanced with multi-output capability, to suggest optimal object handling strategies based on the current scene configuration. The Explainable AI sub-module, implemented using SHAP (SHapley Additive exPlanations), computes feature importance values for each decision and generates human-readable natural language explanations that are presented to users in real time.

3.3.6. Robotic Control and Interaction Layer

At the apex of the architecture lies the robotic control and interaction layer, which translates the structured decisions produced by the reasoning layer into physical and communicative actions. The robotic manipulation sub-module receives target coordinates and action commands, generates smooth motion trajectories using interpolation-based planning algorithms, and executes pick-and-place operations through a servo motor-driven 6-degree-of-freedom robotic arm controlled via an Arduino microcontroller interface. The speech synthesis sub-module converts structured text instructions into spoken output delivered to users through the audio system. The adaptive feedback sub-module monitors the outcomes of robotic actions and user responses, triggering corrective demonstrations and reinforcement messages when discrepancies are detected. The web-based teacher interface sub-module provides real-time monitoring and session management capabilities, enabling educators to oversee and customize the system's educational activities through an intuitive browser-based dashboard.

3.4. Dataset Development and Data Acquisition

High-quality, representative, and diverse training datasets are foundational to the performance of all machine learning components within the integrated system. Given that existing publicly available datasets do not adequately address the specific requirements of the research domains covered by this study, specialized custom datasets were developed for each of the primary classification tasks. The dataset development process was guided by principles of diversity, realism, balance, and rigorous annotation quality, ensuring that the trained models generalize reliably to the real-world deployment conditions encountered during system operation.

3.4.1. Gaze Direction Dataset

A specialized gaze direction dataset was constructed to support the training of the gaze classification model. The dataset encompasses seven distinct gaze direction categories corresponding to the predefined workspace regions within which objects are arranged for robotic interaction. These categories are Upper Left, Upper Center, Upper Right, Lower Left, Lower Center, Lower Right, and Ambiguous/Neutral. The Ambiguous/Neutral category accommodates gaze states that do not correspond clearly to any specific region, reducing the risk of unintended robotic activations caused by ambiguous or transitional gaze positions.

Data collection involved recording facial imagery of multiple subjects while they directed their gaze toward target markers positioned at precise locations within the workspace. The recording protocol incorporated systematic variation across multiple factors to ensure that

the trained model generalizes robustly across the range of conditions encountered during actual system use. Variations included differences in ambient lighting intensity and directionality, head pose orientation and tilt, eye position and orientation relative to the camera, facial appearance across subjects of different ages and ethnicities, and camera angle and distance. These variations were intentionally included to simulate realistic interaction scenarios and minimize the risk of overfitting to controlled laboratory conditions. Following data collection, all images were manually annotated and organized into category-specific folders to support supervised learning.

3.4.2. Educational Object Dataset

A custom object detection dataset was developed to train the YOLO-based object detection module used within the educational interaction component of the system. The dataset consists of images of simple, visually distinct objects selected for their appropriateness in early cognitive learning and pattern recognition tasks for special needs students, including colored balls, boxes, and balloons in multiple colors. Images were captured in real classroom-like environments under varying lighting conditions, object orientations, distances, and background configurations to ensure that the model performs reliably in authentic deployment settings. Each image was manually annotated using the LabelImg annotation tool, with tight bounding boxes drawn around each visible object and class labels assigned from the predefined category set. A second-annotator quality control review was implemented to ensure annotation accuracy and consistency.

3.4.3. Material Classification Dataset

A material texture classification dataset was assembled to train the DINOv2-based classification model. The dataset encompasses five material categories: plastic, glass, rubber, metal, and other. The inclusion of the "other" category enhances the robustness of the classification system by enabling it to handle objects that do not belong clearly to any of the primary categories, reducing the risk of forced misclassification. Images were captured using an IP Webcam-based mobile device setup, providing flexibility in imaging angles and positions. Multiple images of each object were captured from diverse viewpoints and under varying lighting and background conditions to encourage the model to learn intrinsic material characteristics such as surface texture, reflectivity, and transparency rather than superficial visual cues. Variations in object size, shape, and surface condition, including smooth, rough, glossy, and matte finishes, were also systematically incorporated.

3.4.4. Sorting Arrangement Dataset

A specialized dataset for training the predictive robotic sorting recommendation system was constructed, consisting of 2,080 annotated images depicting various arrangements of colored geometric objects on a flat workspace surface. The objects were drawn from nine classes defined by the combination of color (red, green, blue) and shape (box, pyramid, ball), yielding a fine-grained semantic categorization that supports meaningful sorting strategy reasoning. All images were annotated using the Roboflow platform, enabling efficient bounding box labeling and class assignment. Systematic variation in the number of objects, their spatial arrangement, degree of overlap, orientation, lighting conditions, background surface, capture device, and camera distance and angle was incorporated to

ensure comprehensive coverage of the deployment conditions anticipated during system operation.

3.4.5. Dataset Splitting and Balancing

Across all datasets, a standard train-validation-test split ratio of 70%, 15%, and 15% was applied to enable proper model training, hyperparameter optimization, and unbiased performance evaluation. Stratified splitting was implemented to ensure proportional representation of all categories across the three subsets. Class distribution was carefully monitored in each dataset, and where class imbalances were detected, corrective measures were applied, including targeted data collection for underrepresented categories and the application of Synthetic Minority Over-sampling Technique (SMOTE) for tabular feature-based datasets. This attention to class balance is particularly important in the educational and assistive application context, where the system must perform reliably and equitably across all interaction categories to support fair and effective outcomes for all users.

3.5. Data Preprocessing and Augmentation

Comprehensive data preprocessing was applied across all datasets prior to model training to ensure input consistency, optimize image quality, and eliminate sources of variance that could impair model learning and generalization. The preprocessing procedures were tailored to the specific input requirements of each deep learning model while adhering to shared principles of standardization and quality assurance.

Image preprocessing steps applied across datasets included resizing all images to the specific input dimensions required by each model architecture, ensuring uniform input representation throughout the training and inference pipelines. Pixel normalization was

performed to scale image intensity values to the standard range expected by the respective model, facilitating stable gradient computation and faster convergence during training. Face-region alignment and cropping was applied specifically to the gaze direction dataset to isolate the relevant facial regions and reduce the influence of irrelevant background information on classification. Noise reduction techniques were applied where necessary to minimize distortions introduced by sensor artifacts, compression, or adverse lighting conditions.

To enhance model generalization and reduce the risk of overfitting, comprehensive data augmentation was applied during the training phase across all image-based datasets. Augmentation operations were implemented using the Albumentations library, which provides an efficient and flexible augmentation pipeline fully compatible with object detection and classification workflows. The augmentation repertoire included random horizontal flipping, random rotation within defined angular ranges, horizontal and vertical pixel shifts, random scaling and zoom transformations, brightness and contrast adjustments, HSV color space jittering, mosaic augmentation for object detection training, and injection of synthetic noise perturbations. These transformations introduce controlled variability into the training data, simulating the natural range of conditions encountered in real-world deployment and enabling the models to maintain high performance across diverse and previously unseen inputs.

For the tabular feature-based sorting recommendation dataset, preprocessing involved conversion of raw categorical features to numerical representations through one-hot encoding, computation of aggregate features such as total object counts by color and shape, and feature scaling to ensure that all input features occupy comparable numerical ranges. Class imbalance in the sorting strategy target variable was addressed through the application of SMOTE, which generates synthetic samples for underrepresented classes by interpolating between existing samples in feature space, thereby restoring class balance without the information loss associated with random undersampling.

3.6. Vision-Based Perception Modules

3.6.1. Gaze Direction Classification Using ResNet-18

The gaze direction classification model constitutes the perceptual foundation of the gaze-guided human-robot interaction component of the system. Its primary function is to analyze preprocessed facial imagery captured from an RGB camera and assign the user's current gaze to one of seven predefined spatial regions corresponding to areas of the robotic workspace. Accurate and reliable gaze classification is critical to the usability of the assistive interaction framework, as errors in gaze estimation propagate directly to incorrect robotic actions that could frustrate or harm the user.

Convolutional Neural Networks were selected as the model family for gaze classification based on their established effectiveness in image-based classification tasks and their capacity to automatically extract hierarchical visual features relevant to gaze estimation from raw image data without requiring manual feature engineering. The lower layers of CNN architectures extract low-level visual primitives such as edges, lines, and textures, while higher layers progressively aggregate these primitives into increasingly abstract representations capturing gaze direction, head orientation, and facial geometry cues. This hierarchical feature learning is particularly well-suited to the gaze classification task, where the relevant visual patterns span multiple spatial scales and levels of abstraction.

ResNet-18 was selected as the specific CNN architecture for gaze classification following a comparative evaluation against alternative architectures including MobileNet and VGG16. ResNet-18 is a member of the residual network family, featuring 18 trainable layers that leverage residual skip connections to facilitate effective feature learning in deeper networks. The residual learning principle enables each layer to learn a residual mapping relative to its input rather than an unconstrained transformation, expressed formally as $H(x) = F(x) + x$, where $H(x)$ represents the desired output mapping, $F(x)$ represents the residual function learned by the stacked layers, and x represents the identity

input. This formulation allows the network to preserve valuable lower-level features through skip connections while simultaneously learning higher-level discriminative representations relevant to gaze direction classification. Comparative experiments confirmed that ResNet-18 achieved superior overall classification accuracy, higher regional discrimination performance, and greater generalization across diverse gaze classes compared to MobileNet and VGG16, making it the optimal choice for deployment within the integrated system.

The ResNet-18 model was trained using a supervised learning paradigm with the custom gaze direction dataset. The cross-entropy loss function was employed as the training objective for the multi-class classification problem, optimized through backpropagation with carefully tuned hyperparameters including learning rate, batch size, number of training epochs, and optimization strategy. Validation monitoring was conducted throughout training to track generalization performance, and early stopping was applied to terminate training when validation performance ceased to improve, preventing overfitting to the training set. Hyperparameter tuning was conducted through systematic experimentation to identify the parameter configuration that maximized validation classification accuracy.

3.6.2. Object Detection Using YOLO

The object detection module provides the visual perception capability required for the educational interaction and sorting components of the integrated system. The YOLO (You Only Look Once) architecture was selected as the core object detection framework due to its ability to perform real-time multi-object detection with high speed and accuracy. Unlike traditional two-stage detection approaches that perform region proposal and classification as sequential steps, YOLO processes the entire image in a single forward pass through the network, simultaneously predicting bounding box locations and class

probabilities for all objects present. This single-pass processing paradigm is essential for the interactive application context, where immediate detection responses are required to maintain fluid and engaging interaction experiences.

The YOLO model was trained using transfer learning, with model weights initialized from a pre-training on the large-scale COCO dataset encompassing over 200,000 labeled images across 80 object categories. This pre-initialization leverages visual feature representations learned from a diverse and extensive dataset, significantly reducing the amount of task-specific training data required to achieve high detection performance. The pre-trained weights were subsequently fine-tuned on the custom educational object dataset through stochastic gradient descent optimization with a learning rate schedule comprising a warm-up phase followed by cosine annealing, which has been demonstrated to improve convergence stability and final model performance. Hyperparameter optimization was conducted through a systematic grid search over key training parameters.

Following training, the YOLO model is deployed in real time to continuously detect and classify objects placed within the workspace. Detected objects and their spatial positions are processed to generate structured learning sequences for educational activities and to determine appropriate sorting actions. Model quantization and GPU hardware acceleration were applied to optimize inference speed, enabling the system to process video frames at a sufficient rate to maintain responsive and fluid interaction for users.

3.6.3. Material Classification Using DINOv2

The material texture classification module is responsible for distinguishing between four primary material categories — plastic, glass, rubber, and metal — based on fine-grained visual texture analysis. Accurate material classification is challenging due to the visual similarity of certain material pairs under varying lighting conditions and the subtle nature of discriminative texture cues such as surface reflectivity, transparency, and surface finish.

An extensive comparative evaluation of multiple deep learning architectures was conducted to identify the optimal model for this task.

Initial baseline experiments employed a Sequential Convolutional Neural Network, which achieved approximately 44% classification accuracy, demonstrating insufficient capacity for capturing the complex texture patterns required for reliable material discrimination. Subsequently, more advanced CNN architectures were evaluated: EfficientNet achieved approximately 88% accuracy and MobileNet approximately 89%, representing substantial improvements over the baseline but still exhibiting limitations in distinguishing highly similar material textures. To overcome these limitations, DINOv2, a state-of-the-art self-supervised vision transformer, was adopted as the final classification model.

DINOv2 distinguishes itself from CNN-based approaches through its use of a self-supervised learning paradigm that trains the model to produce consistent representations across different augmented views of the same image. This training strategy, grounded in the principles of knowledge distillation between student and teacher networks, enables DINOv2 to learn global contextual representations that capture subtle and long-range spatial dependencies in texture patterns. These global representations provide superior discriminative power for material classification tasks that require distinguishing between materials with visually similar local textures but differing global reflectance and surface properties. DINOv2 achieved approximately 98% classification accuracy on the test set, substantially outperforming all CNN-based alternatives and demonstrating strong robustness to variations in lighting, background, and object orientation.

3.7. Spatial Mapping and Scene Understanding

Spatial understanding of the workspace environment is a prerequisite for translating abstract perceptual outputs into concrete and accurate robotic actions. The integrated system employs two complementary spatial mapping approaches tailored to the requirements of the gaze-guided interaction and object sorting components respectively.

3.7.1. Gaze-to-Region Workspace Mapping

The gaze-to-region workspace mapping module establishes the correspondence between classified gaze directions and target object locations within the robotic workspace. Under this scheme, the workspace is partitioned into six active selection regions — Upper Left, Upper Center, Upper Right, Lower Left, Lower Center, and Lower Right — each corresponding to a physical area where objects may be placed for robotic selection. The mapping between gaze regions and workspace areas is defined by a structured lookup table that assigns each gaze class to a specific target coordinate tuple (x, y, z) in the robot's reference frame. This mapping scheme simplifies the interface between perception and robotic action, enabling the user to issue high-level object selection commands through natural gaze direction without requiring any explicit manual input.

A temporal stability verification mechanism is integrated into the mapping process to mitigate the effects of gaze noise and transient prediction errors. Object selection is only triggered when a consistent gaze region prediction is obtained across a predefined number of consecutive video frames and the prediction confidence score exceeds a calibrated threshold value. This multi-frame consistency requirement ensures that robotic actions are initiated only in response to deliberate and sustained gaze direction, preventing accidental activations caused by brief eye movements or classification uncertainties.

3.7.2. Homography-Based Scene Coordinate Transformation

For the sorting and educational components, a more sophisticated spatial mapping approach is employed to achieve precise object localization within the three-dimensional workspace. The working surface is first detected through edge detection and contour analysis algorithms, and homography transformation is subsequently applied to project the camera's perspective view of the workspace into a normalized top-down coordinate space. This transformation corrects for perspective distortion introduced by the camera's viewing angle, ensuring that the computed positions of objects in the normalized coordinate space correspond accurately to their physical positions in the workspace.

The normalized workspace is subdivided into 22 discrete regions that represent the possible positions where objects may be placed for robotic interaction. Each detected object, after coordinate transformation, is associated with the nearest predefined region, enabling the system to reason about object positions in terms of a discrete spatial grammar that is compatible with the structured action representations used by the robotic control system. This region-based spatial abstraction provides robustness to minor variations in object placement and camera positioning, maintaining reliable spatial reasoning across different session configurations.

3.8. Predictive Sorting and Recommendation System

The predictive sorting recommendation module provides the decision-making intelligence for the robotic sorting components of the integrated system. Unlike rule-based sorting systems that apply fixed, hard-coded sorting criteria, the recommendation module employs data-driven machine learning models that learn optimal sorting strategies from

labeled training examples, enabling the system to generalize to novel scene configurations and adapt recommendations to the active pedagogical mode

3.8.1. Feature Engineering and Data Preparation

Prior to model training, extensive exploratory data analysis was conducted on the sorting dataset to assess data quality, examine class distributions, and identify potential sources of bias. Structured feature engineering was applied to transform raw object detection outputs into a tabular feature representation suitable for machine learning. The feature set comprises object count variables for each combination of color and shape (nine features corresponding to the nine object classes), aggregate summary features such as total object counts by color and by shape, and a one-hot encoded indicator variable representing the active lesson or sorting mode. Categorical variables were converted to numerical representations, and continuous features were scaled to a standardized range for models sensitive to feature magnitude

3.8.2. Random Forest Classifier

The primary machine learning model employed for sorting strategy recommendation is a Random Forest classifier configured for multi-output classification, enabling the simultaneous prediction of multiple sorting targets from a single model inference. The Random Forest architecture aggregates the predictions of 200 individual decision trees, each trained on a different bootstrap sample of the training data, with a maximum tree depth constraint applied to prevent overfitting. The ensemble averaging process reduces prediction variance and improves generalization compared to individual decision trees, while the tree-based structure inherently supports the computation of feature importance

scores that indicate the relative contribution of each input feature to the model's predictions. These importance scores form a natural basis for the explainability layer, enabling the system to identify and communicate the key factors that drove each sorting recommendation.

3.8.3. XGBoost and Logistic Regression Models

To enable rigorous comparative evaluation of the recommendation module's performance, two additional machine learning models were implemented and evaluated alongside the Random Forest approach. An XGBoost gradient boosting model was developed as a high-performance alternative, employing a sequential tree-building process in which each successive tree learns to correct the errors of its predecessors through residual fitting. While XGBoost demonstrated strong predictive accuracy, its sequential boosting process introduces higher computational overhead and reduced interpretability compared to the Random Forest. A Logistic Regression model was implemented as a simple linear baseline, providing a reference point for assessing the performance benefit of the more complex non-linear models. Logistic Regression demonstrated the lowest predictive performance among the three approaches, consistent with expectations given the inherently non-linear relationships between scene configurations and optimal sorting strategies.

3.9. Explainable AI Integration

The Explainable AI module represents a defining distinguishing feature of the integrated system, transforming what would otherwise be an opaque automated decision-making process into a transparent, interpretable, and educationally meaningful interaction. Explainability is not treated as a post-hoc addition to the system but is embedded as a core functional requirement from the earliest stages of the design process. The module is designed to generate accurate, consistent, and comprehensible explanations of the system's sorting recommendations in real time, supporting both user trust and pedagogical effectiveness.

3.9.1. SHAP-Based Model Explanation

The SHAP (SHapley Additive exPlanations) framework is employed as the primary explainability mechanism. SHAP is grounded in cooperative game theory and computes explanation values by determining the marginal contribution of each input feature to a particular model prediction, averaged over all possible subsets of the remaining features. This approach guarantees that the computed explanation values satisfy a set of desirable mathematical properties including local accuracy, missingness, and consistency, ensuring that the explanations faithfully reflect the model's actual reasoning rather than approximating it through a surrogate model. For each sorting recommendation generated by the system, SHAP values are computed for all input features, quantifying how each feature — such as the count of red boxes, the number of blue balls, or the active lesson mode — contributed positively or negatively to the predicted sorting strategy

3.9.2. Natural Language Explanation Generation

To make the SHAP-derived feature contributions accessible to users without technical backgrounds, a natural language explanation generation module was developed to translate numerical SHAP values into coherent, contextually appropriate verbal explanations. The generation process involves ranking the input features by their absolute SHAP contribution magnitudes, selecting the top-contributing features, and mapping the selected features and their contribution directions to predefined explanation templates. The resulting explanations are dynamically constructed based on the specific inputs and outputs of each prediction, ensuring that they accurately reflect the actual reasoning behind each individual recommendation rather than providing generic or pre-scripted responses. For example, if the predominance of red objects is the primary driver of a recommendation to sort by color, the system generates an explanation such as "The sorting recommendation prioritizes color grouping because red-colored objects are the most prevalent category in the current scene."

3.9.3. Visual Explainability and Interactive Transparency

Complementing the natural language explanations, the system provides visual explainability through graphical representations of the current scene's object distribution by color and shape, displayed prominently within the user interface. These visualizations directly correspond to the features considered by the recommendation model, enabling users to verify the connection between the scene state and the system's reasoning. The interface additionally highlights the recommended sorting action and presents the full explanation in a structured layout designed for clarity and accessibility, even for users with limited technical backgrounds. The interactive design allows users to observe in real time how the explanations evolve as the scene configuration changes, reinforcing the

association between scene inputs and recommendation outputs and supporting deeper conceptual understanding of the underlying decision logic.

In the educational context, the explanations are calibrated to align with fundamental learning concepts such as counting, categorization, grouping, and comparison, enabling students to observe the robot's reasoning and reproduce it independently. Teachers are encouraged to use the explanations as instructional tools, prompting students to predict the system's recommendation before it is displayed and then comparing their reasoning to the system's explanation. This pedagogical use of explainability transforms the robotic system from a passive demonstrator into an active reasoning partner that supports the development of logical thinking and pattern recognition skills.

3.10. Speech Recognition and Verbal Interaction Module

The speech recognition and verbal interaction module provides the auditory communication channel through which the system engages with users in the educational interaction component. This module encompasses the complete processing pipeline from audio capture through speech-to-text transcription, text tokenization, verbal response classification, and adaptive feedback generation.

3.10.1. Speech-to-Text Processing Using Whisper

OpenAI's Whisper automatic speech recognition model was selected for speech-to-text processing due to its demonstrated robustness to diverse acoustic conditions, varied pronunciation patterns, and multiple accents. Whisper is a large-scale deep learning model trained on a diverse multilingual audio dataset, enabling it to achieve high transcription accuracy even in the presence of moderate background noise and with speakers of varying

ages and speech clarity. The model processes audio segments captured through the system's unidirectional microphone, which undergoes pre-processing steps including noise reduction, silence removal, and volume normalization before being passed to the recognizer. The transcribed text output serves as the input to the subsequent text processing and evaluation stages of the pipeline.

3.10.2. Tokenization and Response Processing

The continuous text output from the Whisper recognizer is processed through a tokenization pipeline that extracts the structured sequence of object tokens described by the student's verbal response. This pipeline applies word segmentation, stop word removal, color term normalization, and object name matching to identify and extract references to the objects in the current learning sequence. The tokenizer is designed to handle natural variations in student expression, mapping a range of equivalent verbal formulations to the correct canonical object tokens. For example, expressions such as "the red one" and "the big red ball" are both correctly mapped to the canonical token for the red ball object class.

3.10.3. Verbal Response Classification Using Logistic Regression

A logistic regression classifier trained on labeled student response data is employed to evaluate whether the student's verbal response correctly matches the expected learning sequence. The classifier operates on a feature representation that captures key dimensions of response accuracy, including sequence alignment scores, token overlap ratios, order accuracy metrics, and response length difference features. The classifier outputs a probability score for correct classification, with a threshold-based decision rule applied to

generate a binary correct or incorrect judgment. This judgment determines the system's subsequent response: if the student's answer is judged correct, the system advances to the next learning stage with positive reinforcement; if judged incorrect, an adaptive feedback response is generated.

3.10.4. Adaptive Multimodal Feedback Mechanism

The adaptive feedback mechanism is designed to respond in a sensitive and graduated manner to the nature and degree of the student's error, mirroring the scaffolded instruction strategies employed by skilled special needs educators. For responses that are close to correct but contain minor errors, such as a single transposed token or a slightly mispronounced object name, the feedback provides targeted corrective guidance that highlights the specific error without repeating the entire learning sequence. For responses that are substantially incorrect, a more comprehensive feedback response is generated that includes re-demonstration of the full sequence by the robotic arm and repetition of the verbal instruction. This multimodal feedback — delivered simultaneously through both the robotic arm's physical demonstration and the speech synthesis system's verbal output — reinforces the correct learning pattern through complementary visual and auditory channels, improving retention and comprehension.

3.11. Robotic Arm Control Architecture

The robotic manipulation system constitutes the physical action layer of the integrated platform, translating high-level intention signals and decision outputs into precise, safe, and coordinated physical interactions with objects in the workspace. The robotic system is designed to operate reliably and safely in shared human-robot environments, with

particular attention to the safety requirements imposed by the presence of differently-abled users and special needs students in close proximity to the robot during operation.

3.11.1. Hardware Components

The physical manipulation platform consists of a 6-degree-of-freedom (6-DOF) robotic arm assembly that provides sufficient kinematic flexibility to position its end effector at any point within the workspace and at any orientation, enabling a wide repertoire of interaction behaviors including pointing, grasping, lifting, placing, and sweeping. The arm is actuated by MG996R servo motors at each joint, providing the torque, positional accuracy, and stability required for controlled pick-and-place operations. The servo motors are controlled by an Arduino MEGA 2560 microcontroller, which receives motion commands from the host computer via a serial USB communication interface and translates them into pulse-width modulation (PWM) signals through a PCA9685 16-channel PWM driver module. Stable power delivery to all servo motors is ensured through an LM2596 DC-DC buck converter that steps down the 12V supply to the 5V level required by the actuators, preventing voltage-induced jitter and ensuring consistent actuation performance.

3.11.2. Motion Planning and Trajectory Generation

Motion planning within the robotic control system is responsible for generating smooth, safe, and accurate joint trajectories that guide the robot from its current configuration to the target configuration required for each manipulation task. The trajectory generation algorithm produces interpolated waypoint sequences between the current and target joint angles, yielding fluid and continuous movements that are intuitively comprehensible to

student observers and mechanically safe for the servo actuators. Velocity and acceleration constraints are applied to all generated trajectories to ensure that joint movements remain within the safe operating range of the servo motors, preventing mechanical stress and potential hardware failures. Collision avoidance constraints are enforced to prevent the arm from contacting objects on the workspace surface or the hands of students during interaction.

3.11.3. Pick-and-Place Execution Pipeline

The pick-and-place execution pipeline implements the complete sequence of robotic actions required to locate, grasp, transport, and place an object in response to a gaze selection, learning demonstration, or sorting decision. The pipeline is decomposed into a structured sequence of phases. Initially, the robotic arm moves from its home position to a pre-grasp pose positioned above the target object at a safe approach distance. The end effector is then aligned with the target object using the object's localized position coordinates. Grasp execution is performed when target localization is confirmed, closing the gripper around the object. The object is lifted vertically to a safe transport height, transported horizontally to the destination position, and carefully placed. The gripper is then released, and the arm returns to its home position in preparation for the next interaction cycle. This structured pipeline ensures predictable, safe, and repeatable manipulation behavior across diverse object positions and target destinations.

3.11.4. Interactive Learning Workflow Integration

In the educational interaction context, the robotic manipulation pipeline is extended with an interactive learning phase specifically designed to transform the sorting process into an engaging pedagogical experience. Following initial object pickup and placement within a designated Region of Interest for visual classification, the robotic arm moves the classified object to a front-facing position accessible to the student, pausing to allow the student to observe the object closely and respond to a classification prompt. After the student's response has been evaluated, the arm proceeds to execute the final sorting action, placing the object in the appropriate category bin. This interactive workflow embeds the student's active participation into the robotic task execution sequence, creating a collaborative learning experience in which the student is not merely an observer but an active partner in the classification and sorting process.

3.12. System Integration and Interaction Workflow

The complete integrated system is implemented as a software-hardware co-designed pipeline in which all perception, reasoning, and action components operate in coordinated real time. The implementation is realized primarily in Python for the software processing pipeline, with Arduino-based firmware handling low-level motor control on the hardware side. The system follows a multi-threaded architecture, with separate threads dedicated to video frame capture and processing, speech recognition monitoring, robotic arm command dispatch, and user interface updates. Thread-safe message-passing mechanisms coordinate the concurrent processes, preventing race conditions and ensuring consistent data flow between components.

The complete system interaction logic is formalized as a state machine with clearly defined states and transition conditions. The primary states through which a typical interaction session progresses are: system initialization, user gaze acquisition and classification, intention mapping and confidence validation, target coordinate computation, robotic action planning and execution, speech instruction delivery, student verbal response collection, response evaluation and feedback generation, and session completion or next-cycle initialization. Transitions between states are event-driven, triggered by the completion of robotic motion sequences, the detection of valid student speech input, the expiration of response timeouts, or the satisfaction of confidence threshold criteria. This state machine architecture provides a clear, predictable, and auditable structure for the interaction workflow, ensuring consistent system behavior across different sessions and user profiles.

The overall system operation can be summarized through the following high-level processing pipeline:

Table 1: Workflow

Step	**Process**	**Description**
1	Image and Audio Capture	Acquire facial frames via RGB camera and audio via microphone for gaze and speech analysis.
2	Preprocessing	Crop, resize, normalize facial and audio inputs; apply face detection and noise reduction.
3	Gaze Classification	Classify user gaze region using ResNet-18; detect workspace objects using YOLO; classify material using DINOv2.
4	Confidence Validation	Apply temporal consistency checks and confidence threshold filtering across multiple frames.

5	Intention Mapping	Map classified gaze or detection outputs to workspace coordinates and target object locations.
6	Recommendation and Explanation	Generate sorting recommendation using Random Forest; compute SHAP explanations; produce natural language output.
7	Robot Action Planning	Translate target coordinates into motion trajectories; plan pick-and-place sequence.
8	Execution and Interaction	Execute robotic manipulation; deliver verbal instructions; collect and evaluate student response.
9	Feedback and Reinforcement	Provide adaptive multimodal feedback; reinforce correct responses; re-demonstrate incorrect sequences.
10	Object Handover or Placement	Complete object delivery to user or sorting destination; return arm to home position.

Error handling and recovery mechanisms are implemented throughout the system to manage potential component failures and unexpected environmental events. If the gaze classification model fails to produce a prediction above the confidence threshold, the system remains in the awaiting-input state and provides a visual prompt to the user indicating that their gaze has not been recognized. If the speech recognition module fails to produce a valid transcription, a verbal prompt requests the student to repeat their response. If the robotic arm encounters an unexpected obstruction during motion execution, an emergency halt is triggered and a warning message is generated for the teacher. These recovery mechanisms ensure that the system maintains its functional and educational effectiveness even in the presence of technical challenges or adverse environmental conditions.

3.13. Web-Based Teacher Interface and Support System

A comprehensive web-based teacher interface was developed to support educators in managing, monitoring, and customizing learning sessions within the integrated system. The interface was designed in close consultation with special education teachers to ensure that it meets their practical needs while remaining accessible to users with limited technical training. The interface is implemented using standard web technologies including HTML5, CSS3, and JavaScript, with a responsive design approach ensuring compatibility across desktop computers, tablets, and smartphones. A lightweight backend server provides data management and communication with the system's core processing components.

The main dashboard of the teacher interface provides a real-time overview of the current session, including a live video feed from the workspace camera, a display of the most recently detected and classified objects, the current gaze classification result and confidence level, and a summary of the student's performance metrics including accuracy rate, number of correct and incorrect responses, and average response time. This real-time visibility enables teachers to monitor the interaction closely and intervene if necessary, while remaining in the background when the student is successfully engaging with the system independently.

The learning content management module allows teachers to configure session parameters including the selection of active object categories, the desired sequence length, the difficulty level, and the time allowed for student responses. Difficulty adjustments can be made by varying the number and similarity of objects in the sequence, enabling teachers to tailor the challenge level to the specific abilities and goals of individual students. Historical performance data is accessible through a dedicated progress tracking module that displays session-by-session performance trends, enabling teachers to identify areas of

strength, track learning progress over time, and make evidence-based decisions about instructional planning.

Security and privacy considerations were incorporated throughout the interface design. All data transmitted between the interface and the backend server is encrypted using industry-standard protocols. Student performance data is stored locally on the school's network infrastructure to ensure compliance with data protection regulations and maintain institutional control over sensitive student information. Access to the interface is protected by password authentication to prevent unauthorized access.

3.14. Experimental Setup and Performance Evaluation

System performance was evaluated through a comprehensive multi-stage testing protocol that assessed each component individually and the complete integrated system collectively. The testing methodology follows established practices in intelligent systems evaluation, combining quantitative performance metrics with qualitative assessments of usability, responsiveness, and educational effectiveness.

3.14.1. Gaze Classification Evaluation

The gaze direction classification model was evaluated using standard image classification performance metrics computed on the held-out test set. The primary evaluation metric is overall classification accuracy, computed as the proportion of test samples correctly classified across all seven gaze direction categories. Additional metrics computed include per-class precision, recall, and F1-score, providing a nuanced assessment of the model's

performance across individual gaze categories and revealing any systematic classification biases. Confusion matrix analysis was conducted to visualize the pattern of misclassifications across categories, enabling the identification of gaze direction pairs that exhibit higher confusability and informing potential refinements to the training data or model architecture.

3.14.2. Object Detection and Material Classification Evaluation

The YOLO object detection model was evaluated using precision, recall, and mean Average Precision (mAP) metrics computed across all object categories on the held-out test set. A controlled test protocol was implemented that systematically varied environmental parameters including lighting intensity, object distance, background complexity, and object orientation, enabling analysis of the model's performance across different deployment conditions. The DINOv2 material classification model was evaluated using overall accuracy, per-class precision, recall, and F1-score metrics. The superior performance of DINOv2 relative to the baseline CNN, EfficientNet, and MobileNet architectures was confirmed through controlled comparative experiments using identical training and evaluation datasets.

3.14.3. Speech Recognition and Classification Evaluation

The Whisper speech recognition module was evaluated using the Word Error Rate (WER) metric on a dedicated test dataset of verbal responses representative of the target student population. The dataset encompassed responses of varying pronunciation clarity, speech rate, and ambient noise level to assess recognition robustness across realistic deployment conditions. The logistic regression verbal response classifier was evaluated using

precision, recall, and F1-score metrics computed on the labeled response evaluation dataset. Usability of the teacher interface was assessed through structured testing sessions with experienced special education teachers using the System Usability Scale (SUS) questionnaire, supplemented by think-aloud observations and post-task interviews.

3.14.4. Sorting Recommendation Evaluation

The sorting recommendation models were evaluated using classification accuracy and F1-score metrics on the held-out test split of the sorting dataset. The Random Forest, XGBoost, and Logistic Regression models were evaluated under identical conditions to enable fair comparative assessment. Model robustness was assessed by evaluating performance across scene configurations of varying complexity, including scenes with different numbers of objects, varying degrees of class imbalance, and novel arrangements not represented in the training data. XAI explanation quality was assessed qualitatively through structured review sessions with domain experts and educators, evaluating the accuracy, comprehensibility, and educational relevance of the generated natural language explanations.

3.14.5. Task-Level and System-Level Evaluation

Beyond component-level metrics, the complete integrated system was evaluated at the task level through a series of end-to-end interaction trials conducted in a simulated classroom environment. Task-level metrics assessed included object selection success rate for the gaze-guided interaction component, defined as the proportion of trials in which the robotic system correctly selected and retrieved the object corresponding to the user's intended gaze region; pick-and-place execution success rate, defined as the proportion of

robotic action sequences completed without error or intervention; system responsiveness, assessed qualitatively through observation of interaction fluency and the absence of perceptible delays between user input and robotic response; and interaction reliability, defined as the consistency of correct task completion across repeated trials under varied conditions.

3.15. Safety and Reliability Considerations

Given that the integrated system operates in shared environments with human users, including individuals with motor and communication impairments and children with special needs, safety and reliability considerations were treated as primary design constraints throughout the development process. Multiple complementary safety mechanisms were incorporated to minimize the risk of harmful robotic actions, false activations, and system failures.

At the perception and decision-making level, confidence thresholding ensures that robotic actions are only initiated when classification predictions meet or exceed calibrated minimum confidence levels. Multi-frame temporal consistency verification requires that gaze predictions remain stable across a predefined number of consecutive frames before triggering selection, preventing accidental activations due to transient noise or eye movements. At the robotic control level, joint movement limits are implemented as software constraints that prevent the arm from exceeding its safe operating range. Smooth motion trajectories generated by the interpolation-based planning algorithm minimize dynamic loads on mechanical joints and reduce the risk of abrupt movements that could startle or harm nearby users. Collision avoidance constraints prevent the arm from entering regions of the workspace where users' hands or other objects may be present during interaction.

System-level reliability was enhanced through modular testing of each component prior to integration, enabling the early identification and resolution of component-level failures before they could propagate to the integrated system. Comprehensive error handling and recovery mechanisms ensure that the system responds gracefully to unexpected events without requiring manual intervention in most cases. Fail-safe stopping criteria are implemented at all levels of the system, ensuring that any detected anomaly triggers an immediate halt of robotic action and notification to the supervising teacher. These safety and reliability measures are particularly critical in the assistive and educational contexts targeted by this research, where the consequences of system errors could have significant implications for user wellbeing and trust

3.16. Commercialization Potential and Applications

The integrated human-robot collaboration platform developed in this research exhibits substantial commercialization potential across multiple application domains, driven by its combination of technological innovation, accessibility-oriented design, cost-effective hardware implementation, and broad applicability in healthcare, rehabilitation, and educational markets. By leveraging inexpensive computer vision and deep learning components in place of costly proprietary sensing hardware, the system offers a commercially attractive alternative to existing assistive robotic solutions, many of which rely on specialized equipment that places them beyond the financial reach of many potential users and institutions, particularly in developing regions.

In the assistive technology market, the platform addresses a growing demand for intelligent systems that enable users with motor disabilities to perform daily tasks independently. The gaze-guided object selection capability enables users to interact with robotic workstations using only eye direction, without requiring any manual dexterity or physical effort. Potential deployment contexts include residential settings, rehabilitation

institutions, assisted living facilities, and special education centers. Specific use cases encompass medication retrieval, classroom assistance, object passing, and other tasks requiring robotic mediation for physically impaired individuals.

In the educational technology market, the interactive learning and material classification components offer a novel approach to inclusive and experiential learning that aligns with the growing trend toward technology-enabled differentiated instruction. The system's ability to adapt task difficulty, provide multimodal feedback, and generate transparent AI-driven explanations makes it particularly well-suited for deployment in special needs educational environments where engagement, accessibility, and individualization are paramount. The explainability features additionally provide value in mainstream educational settings as tools for teaching logical reasoning, pattern recognition, and AI literacy concepts.

The software-centric design of the system's core intelligence enables flexible commercialization strategies beyond pure hardware sales. The technology may be packaged and commercialized as a gaze-guided assistive robotic system for direct sale to end users and institutions; as a software licensing offering for integration into third-party robotic platforms; as an institutional subscription service providing cloud-managed learning session analytics and content management; and as a customizable deployment solution for specialized applications in healthcare, rehabilitation, and industrial collaborative robotics. This multi-channel commercialization strategy enhances revenue potential while maximizing market reach across different customer segments and purchasing capabilities.

Future development directions that would strengthen the commercialization position of the platform include integration of multimodal control modalities such as voice commands and gesture recognition to complement gaze-based interaction, extension of the educational content library to support a broader range of learning objectives and curriculum areas, development of home automation integration capabilities to extend the system's assistive reach beyond structured workstation settings, and implementation of

adaptive personalization algorithms that tailor the system's behavior to individual user profiles based on accumulated interaction history. Intellectual property protection through patents covering novel gaze-directed robotic decision algorithms, region-intention mapping techniques, and XAI-integrated assistive system architectures would further enhance long-term commercial viability.

4. Results and Discussion

4.1. Results

The texture recognition module was implemented with the aid of a self-supervised vision transformer DINOv2 fused with a customized PyTorch classifier model, which was trained using a dataset that included Glass, Metal, Plastic, Rubber, and Others. In order to assess the performance of the proposed method, other model architectures like Sequential CNN (Accuracy = 0.4419), EfficientNet (Accuracy = 0.8772), and MobileNet (Accuracy = 0.8872) were considered for comparison.

It is evident from Table 4 that DINOv2 outperformed all other baseline methods by a wide margin, with a test accuracy of 0.98, and precision, recall, and F1-scores ranging from 0.95 to 0.99 in all categories. More insights into the effectiveness of the model in discriminating between different materials based on their features can be seen from the t-SNE plots where it is clearly observable that all the embeddings of features belonging to a certain material class have been separated from other material classes despite differences in light conditions, orientation, and texture.

The inference on unknown data yielded high confidence scores greater than 0.95, showing that this is indeed a reliable model for real-world applications. Using OpenCV, NumPy, DINOv2, and PyTorch for classification is highly efficient and interpretable.

Pattern learning through vision is performed through the use of YOLO object detection to ensure real-time object location and mapping of objects into predefined spatial areas. By detecting various objects within the environment, patterns are generated through a frequency-based matching method that will work for the detected objects, thus moving away from predefined fixed patterns toward contextually aware patterns. The generated

pattern is partially shown to the user, while the last element is purposely concealed to generate a pattern completion problem.

The robotic system performs the task in steps to help users learn in a guided manner. Interaction within the robotic system is captured in a state-based approach that ensures actions of the robot and reactions of the user are recorded to create a learning loop in real time. In order to classify the reaction of users, Logistic Regression classifier was used, and its accuracy is 0.96, indicating reliable classification of interactions.

The performance of the gaze classification module was assessed on the basis of two models namely ResNet-18 architecture and MobileNet based architecture. Using ResNet-18 architecture, a test accuracy score of 86.50% with macro F1-score value equal to 0.8643 and a weighted F1-score of 0.8657 was obtained, thereby proving its consistent performance for all gaze categories. Excellent classification accuracy scores were obtained for well-defined gaze directions such as upper right (F1=0.9565) and center (F1=0.9500), which show the capacity of the proposed system to detect the gaze pattern effectively. For the ambiguous category of gaze, a relatively low F1 score (F1=0.7568) was obtained, which indicates that this type of gaze cannot be effectively classified.

On the other hand, MobileNet showed much lower performance values in terms of accuracy (33%) and macro F1 score (0.34). Such inferior performance can be attributed to the compact nature of the network architecture. The results suggest that ResNet-18 architecture offers much better feature extraction and classification ability compared to the other network architecture. Therefore, it can be used effectively for human-robot interaction applications based on intention-driven approach through gaze mapping to the predefined spatial location.

The object detection component, designed based on YOLOv8, is crucial in providing effective environmental perception. It has obtained scores of 0.949 for precision, 0.956 for recall, and 0.954 for mAP@50. This shows the model has performed well in detecting different objects including balls, boxes, and pyramids. Even though the model is relatively moderate when it comes to the mAP@50-95 criterion, with a score of 0.543, its performance is adequate for use in practical real-world robotic systems which do not need very precise location information. Furthermore, the model's quick inference time (2.8 ms per frame) and small size (6.3 MB) make it feasible for deployment. After successful object detection, several models were compared in terms of sorting recommendation.

The accuracy of the XGBoost model was 0.88, whereas Logistic Regression had the same score. In contrast to them, the Random Forest model performed better with about 0.94 accuracy that did not differ much regardless of the number of output classes (color - red, green, blue; shape – ball, box, pyramid). The use of a Multioutput Classifier was important in order to implement multi-target classification in the model. It allows for implementing the mode-based approach to sorting that considers frequency as a key criterion and improves overall efficiency. The proposed design involves the fusion of YOLOv8 based object detection approach with the random forest recommendation algorithm in real time via the use of an IP camera and a robotically controlled setup using an Arduino device. Objects detected in real time are aligned in a grid pattern and undergo a process flow which sorts them and interacts with them according to specific criteria. The implementation of explainable AI via SHAP enables one to interpret the decision process used by the algorithm in an attempt to help educators understand the model and allow students to sort, compare and identify objects.

Table 2: Results and Findings

Component	Models Evaluated	Key Metrics	Selected Model
Texture Recognition	**DINOv2 (0.98),** Sequential CNN (0.4419), EfficientNet (0.8772), MobileNet (0.8872)	Accuracy: 0.98, Precision: 0.98, Recall: 0.98, F1-score: 0.98	DINOv2
Gaze Tracking	**ResNet-18 (0.865),** MobileNet (0.33)	Accuracy: 0.865, Loss: 0.7601, Macro F1: 0.8643, Weighted F1: 0.8657	ResNet-18
XAI based Adaptive Recommendation Sorting	(Object Detection) **YOLOv8**	Precision: 0.949, Recall: 0.956, mAP@50: 0.954, mAP@50–95: 0.543	YOLOv8
	(For Recommendations) **Random Forest (0.94),** XGBoost (0.88), Logistic Regression (0.88)	Accuracy: 0.94, Precision: 0.93, Recall: 0.74, F1-score: 0.82	Random Forest
Pattern Recognition	**Logistic Regression (0.9123),** Naive Bayes (0.88), decision tree (0.64)	Accuracy: 0.9123 Precision: 0.8750 Recall: 0.8200 F1-Score: 0.8467	Logistic Regression

Final working system looks like:

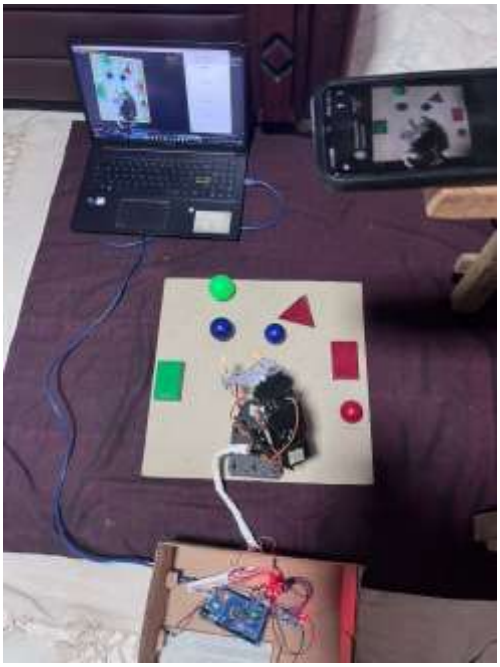

Figure 1. 2: Final working system

4.2. Research Findings

Collectively, all the experimental studies mentioned above have shown that combining computer vision techniques, machine learning algorithms, multimodal sensors, and robotic control leads to the creation of extremely efficient intelligent systems which can be used for such purposes as real-time perception, decision-making, and interactive learning. Modern deep learning techniques perform much better compared to classical ones when it comes to accuracy and robustness, especially in complex real-world conditions featuring a variable level of illumination, diverse appearances of objects and their arrangement.

An important aspect to be mentioned is the fact that, in the vast majority of cases, advanced deep learning architectures perform far better compared to conventional ones such as CNNs. Transformer-based architectures, which operate using self-supervised models, exhibit high competence in detecting both local and global correlations between elements of visual scenes. Thus, they are quite efficient at recognizing materials by analyzing their textures. Conversely, the main limitation of conventional CNN architectures is that they can cope with simple and basic features but not with complex visual scenes exhibiting subtle differences between classes.

YOLO family approaches show consistent efficiency and accuracy in real-time object detection, which confirms that it can be successfully utilized in time-dependent tasks such as robotic sorting and educational interactive activities. The ability of these algorithms to withstand changes in external conditions increases their reliability for further application. Likewise, regression analysis and ensemble methods such as Logistic Regression, Random Forest, and XGBoost prove to have predictive power, while ensemble models

usually provide higher stability and better results than linear models or probabilistic analysis.

The second finding concerns the considerable efficiency gains resulting from multimodality. Combining vision with other types of sensors like ultrasonic sensing, voice recognition, and robotic mechanisms allows one to achieve greater reliability of the system and decrease dependence on one specific type of data. While vision-based models are vulnerable to any possible interference in the environment, the fusion of sensory data provides a certain level of verification and guarantees a more stable decision-making process.

Interpretable artificial intelligence is significant in closing the gap between AI algorithmic processes and human comprehension. The use of interpretability methods ensures clear depiction of prediction formation processes, enhancing trust in AI-enabled applications, especially in academic settings. The feature importance ranking and decision explanation process helps ensure clarity of machine learning algorithms' outputs, strengthening their usability in academic contexts.

Different implementations of robotic actuator systems have shown consistent performances in performing structured tasks such as manipulating objects, sorting, and directing interactions. Microcontroller-based systems integrated into the AI perception pipeline provide seamless coordination between software intelligence and hardware. However, performance depends on the level of mechanical tuning and environmental factors, implying that the physical system design limits functionality more than the computerized component.

The first aspect that stands out from all researches reviewed here is that these kinds of systems are extremely effective educational tools. Interactive work of humans with robots significantly increases engagement and concentration of learners while contributing to

active education through various senses, namely, visual, auditory, and touch. Systems of this kind are especially valuable in assistive education due to multimodality and feedbacks providing additional information, which helps in understanding the material and reduces frustration levels of students. Furthermore, the cost-effectiveness of such technology, achieved by inexpensive devices and open-source software, proves that even highly advanced systems of educational robots can be created without any necessity in expensive equipment.

Nevertheless, some limitations related to small sample sizes of datasets, sensitivity to changes in the environment, increased computation costs of well-performed models, and lack of scalability in dynamic multi-object settings are worth mentioning.

Overall, results presented above prove that the combination of advanced deep learning algorithms, multimodal data processing, explainable AI, and robotics control form an ideal basis for highly intelligent and efficient systems.

4.3. Discussion

The general conclusion about the results obtained shows that the combination of computer vision, machine learning, explainable AI, multimodal sensors, and robotic actuator control provides the necessary basis for creating intelligent systems that perceive the world, reason about it, and interact physically. In all papers considered, advanced deep learning techniques showed better results than conventional algorithms, especially in the case of visual perception, fine-grained classification, and dynamic situations. Transformer-based models and YOLO-based detectors showed great flexibility and accuracy, while classic CNN neural network structures and other types of machine learning models were less efficient under varying circumstances.

One of the main conclusions drawn from this work is that multimodal systems outperform systems with a single modality. The ability of multimodal perception to recognize objects more accurately stems from the fact that single-modality systems have increased sensitivity to various factors, including environmental factors. Multimodal systems not only provide multiple data sources, which reduces error, but also validate decisions based on multiple sources, resulting in better overall performance. Such systems perform better in interaction with human operators as well.

The other crucial point in this regard is the importance of explainability of decisions taken by an AI-powered mechanism in educational or assistive environments. Explainability refers to enabling users to comprehend the underlying decision-making processes used by an AI-powered machine, which enhances transparency and increases trust in the system. The point is quite valid in case of using the system for educational purposes as the process of teaching depends to a significant extent on the ability to provide explanations.

In terms of robotic applications of AI, the integration of an AI-powered model with hardware execution mechanisms works well when it comes to performing tasks in real-time, including the detection of objects, their classification and sorting, as well as guiding them. It should be noted that, despite the successful operation, the efficiency of work largely depends on the level of mechanical tuning as well as on the environment.

One common cross-cutting observation relates to the high applicability of the studied systems within the field of education and assistance. The use of interactive human-robot systems can contribute significantly to increased engagement, attention, and interaction through the implementation of visual, auditory, and physical learning methods. The latter allows for improved comprehension and memory of new information, especially when working with inclusive education systems requiring adaptable and interactive teaching strategies. Furthermore, the possibility of constructing an interactive system based on

relatively cheap hardware and open-source software suggests significant prospects for widespread utilization of the proposed technologies in different conditions.

Nevertheless, certain limitations persist throughout all research studies. In particular, reliance on data size and variety, susceptibility to environmental factors such as lighting and noise levels, computational complexity of high-performance machine learning algorithms, and difficulties with extrapolation to dynamic or multi-object environments are considered major issues.

5. Conclusion and Future works

The research conducted confirms that the fusion of computer vision, machine learning, explainable artificial intelligence, multimodal sensing, and robotic control can become effective tools for the development of intelligent systems capable of providing real-time perception, decision-making, and interactive execution. For all of the proposed frameworks, it was shown that the modern techniques using transformers or YOLO object detectors significantly outperformed classical CNNs and machine learning algorithms in terms of accuracy of predictions. Moreover, it has been proven that an ensemble of models yields better results compared to individual algorithms and deep residual networks prove their effectiveness in feature learning for appearance-based perception.

Finally, it is vital to highlight that multimodal systems proved to yield better results than unimodal systems. Indeed, a combination of vision sensing along with other types of sensing like ultrasonic sensors, speech processing algorithms, and robotic systems proved to be more robust and reliable compared to systems operating based on only vision data.

It allowed reducing uncertainty of perception process and improving decision-making procedure by means of validating decisions. In turn, adding an explainable artificial intelligence component into systems improved interpretability, thus, making AI-based decisions transparent and easier to understand.

From a perspective of robotic implementation, the study proves that AI-based perception models could be successfully integrated into robotic actuators to perform tasks such as object identification, categorization, separation, and interaction in a coordinated manner. Nevertheless, hardware specifications, calibration precision, and external environment factors affect the overall performance of the system; thus, future developments should consider both physical and technological advancements.

The educational relevance of the presented technology is regarded as a notable contribution to the field. It is demonstrated that the involvement of an additional sensory modality improves user engagement and concentration during lessons. In this context, interactive collaboration between robots and humans becomes crucial for providing multimodal learning experiences, which is vital in assistive learning scenarios. Moreover, affordable hardware components and open-source software make the proposed system viable for implementing in both developing and advanced educational settings.

.Further research could be directed at achieving greater efficiency from the developed models that allow using them in mobile devices without losing their high performance and high accuracy level. The collection of more samples, especially from a variety of objects, environments, and actions of the user, would be beneficial for achieving better generalization results and ensuring the system robustness. Moreover, more adaptive multimodal fusion techniques could be applied to improve the responsiveness of the system under dynamically changing conditions. Additionally, development of a multi-robot collaborative, autonomous and adaptive learning system would also help increase its effectiveness.

At the same time, the significance of this research topic is highlighted by the fact that AI and robotic systems are usually created with typical environments and typical users in mind. Thus, there is no consideration of accessibility issues, adaptability, and interactions between the system and people with various abilities. Instead, this research proposes focusing on the socially oriented usage of technology with the special consideration of its usability and applicability to a wide range of learners.

From the perspective of assistive education, the study shows that the development in data science and robotics can generate significant societal benefit if focused on application areas that have not received sufficient attention yet. Apart from the automation and improvement from the technology side, the contribution made by the research is valuable in terms of fostering cognitive processes and improving learning experiences.

In general, the research reflects the shift from the technology-focused approach to a human-centered one. Through the use of machine learning, explainable intelligence, and robotics in the field of education, the system goes beyond simple automation and gains social significance.

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